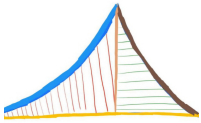
Math for EE



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

CONTENTS

1 CBSE 2019 4
1.1 Class-10 4
1.2 Formula-10 26
1.3 Class-12 30
1.4 Formula-12 49

1. CBSE 2019

A. Class-10

1.1.1 Write the coordinates of the point P on the x-axis which is equidistant from the points $A\left(-2,0\right)$ and $B\left(6,0\right)$. (10 2019)

Solution: Any point on the x-axis has its y-coordinate equal to 0. Let the position vectors of the points be

$$\mathbf{P}=\begin{pmatrix}x\\0\end{pmatrix},\quad \mathbf{A}=\begin{pmatrix}-2\\0\end{pmatrix},\quad \mathbf{B}=\begin{pmatrix}6\\0\end{pmatrix} \tag{1.1.1.1}$$

Since $\mathbf{P}$ is equidistant from $\mathbf{A}$ and $\mathbf{B}$, from (1.2.1.1):

$$\left\|\mathbf{P}-\mathbf{A}\right\|^2=\left\|\mathbf{P}-\mathbf{B}\right\|^2 \tag{1.1.1.2}$$

From (1.2.1.2), evaluating the squared Euclidean distances:

$$\left(x-(-2)\right)^2+\left(0-0\right)^2=\left(x-6\right)^2+\left(0-0\right)^2 \tag{1.1.1.3}$$

$$\left(x+2\right)^2=\left(x-6\right)^2 \tag{1.1.1.4}$$

$$x^2+4x+4=x^2-12x+36 \tag{1.1.1.5}$$

$$16x=32\implies x=2 \tag{1.1.1.6}$$

Therefore, the coordinates of point P are

$$\mathbf{P}=\begin{pmatrix}2\\0\end{pmatrix} \tag{1.1.1.7}$$

The following code verifies the solution in Fig. 1.1.1.1:

codes/10/q1.py

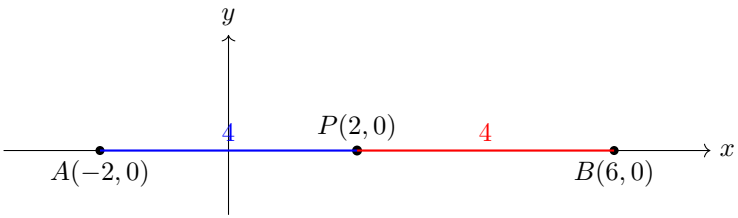


Fig. 1.1.1.1: Point $P(2,0)$ equidistant from $A(-2,0)$ and $B(6,0)$

1.1.2 Evaluate:

$$\sin^2\left(60^\circ\right)+2\tan\left(45^\circ\right)-\cos^2\left(30^\circ\right) \tag{1.1.2.1}$$

(10 2019)

Solution: From (1.2.2.1) and (1.2.2.2):

$$\sin\left(60^\circ\right)=\frac{\sqrt{3}}{2},\quad \tan\left(45^\circ\right)=1,\quad \cos\left(30^\circ\right)=\frac{\sqrt{3}}{2} \tag{1.1.2.2}$$

Substituting these values into the expression:

$$\sin^2(60°) + 2\tan(45°) - \cos^2(30°) = \left(\frac{\sqrt{3}}{2}\right)^2 + 2(1) - \left(\frac{\sqrt{3}}{2}\right)^2 \quad (1.1.2.3)$$

$$= \frac{3}{4} + 2 - \frac{3}{4} = 2 \quad (1.1.2.4)$$

1.1.3 If $\sin A = \frac{3}{4}$, calculate $\sec A$. (10 2019)

Solution: From (1.2.3.2):

$$\cos A = \sqrt{1 - \sin^2 A} = \sqrt{1 - \left(\frac{3}{4}\right)^2} = \sqrt{1 - \frac{9}{16}} = \frac{\sqrt{7}}{4} \quad (1.1.3.1)$$

From (1.2.3.3):

$$\sec A = \frac{1}{\cos A} = \frac{4}{\sqrt{7}} \quad (1.1.3.2)$$

1.1.4 In Fig. 1.1.4.1, ABC is an isosceles triangle right-angled at C with $AC = 4$ cm. Find the length of AB. (10 2019)

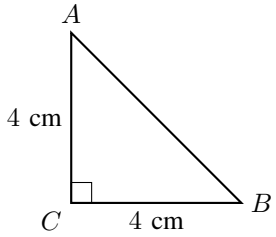


Fig. 1.1.4.1: Isosceles right triangle ABC

Solution: Triangle ABC is an isosceles right-angled triangle at C. Since the hypotenuse AB is the longest side, the two equal legs are AC and BC:

$$BC = AC = 4 \text{ cm} \quad (1.1.4.1)$$

Applying the Pythagoras theorem (1.2.4.1) and (1.2.4.2):

$$AB^2 = AC^2 + BC^2 = 4^2 + 4^2 = 32 \quad (1.1.4.2)$$

$$\implies AB = \sqrt{32} = 4\sqrt{2} \text{ cm} \quad (1.1.4.3)$$

1.1.5 In Fig. 1.1.5.1, $DE \parallel BC$. Find the length of side AD, given that $AE = 1.8$ cm, $BD = 7.2$ cm, and $CE = 5.4$ cm. (10 2019)

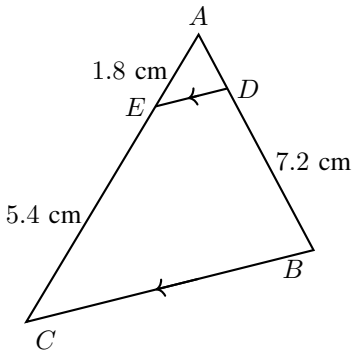


Fig. 1.1.5.1: Triangle ABC with $DE \parallel BC$

Solution: In $\triangle ABC$, since $DE \parallel BC$, applying Thales' Theorem (1.2.5.1):

$$\frac{AD}{BD} = \frac{AE}{CE} \tag{1.1.5.1}$$

Substituting the given lengths:

$$\frac{AD}{7.2} = \frac{1.8}{5.4} = \frac{1}{3} \implies AD = \frac{7.2}{3} = 2.4 \text{ cm} \tag{1.1.5.2}$$

1.1.6 Find the nature of roots of the quadratic equation $2x^2 - 4x + 3 = 0$. (10 2019)

Solution: Comparing with the standard quadratic form $ax^2 + bx + c = 0$:

$$a = 2, \quad b = -4, \quad c = 3 \tag{1.1.6.1}$$

From (1.2.6.1), the discriminant D is:

$$D = b^2 - 4ac = (-4)^2 - 4\,(2)\,(3) = 16 - 24 = -8 \tag{1.1.6.2}$$

Since $D < 0$, the quadratic equation has no real roots. The following code verifies the solution in Fig. 1.1.6.1:

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codes/10/q6.py
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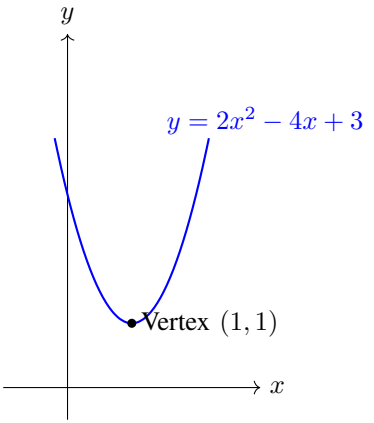


Fig. 1.1.6.1: Plot of $y = 2x^2 - 4x + 3$ showing no real roots ($D < 0$)

1.1.7 If $\gcd(336, 54) = 6$, find $\text{lcm}(336, 54)$. (10 2019)

Solution: Let $a = 336$ and $b = 54$. From (1.2.7.1):

$$\gcd(a, b) \times \text{lcm}(a, b) = a \times b \tag{1.1.7.1}$$

Rearranging:

$$\text{lcm}(336, 54) = \frac{a \times b}{\gcd(a, b)} \tag{1.1.7.2}$$

$$= \frac{336 \times 54}{6} = 336 \times 9 = 3024 \tag{1.1.7.3}$$

1.1.8 Find the 21st term of the A.P. $-4\frac{1}{2}, -3, -1\frac{1}{2}, \ldots$ (10 2019)

Solution: The first term is $a = -\frac{9}{2}$. The common difference d is:

$$d = -3 - \left(-\frac{9}{2}\right) = -3 + \frac{9}{2} = \frac{3}{2} \tag{1.1.8.1}$$

From (**??**), the n-th term is:

$$a_n = a + (n - 1)\, d \tag{1.1.8.2}$$

Substituting $n = 21$:

$$a_{21} = -\frac{9}{2} + (21 - 1)\left(\frac{3}{2}\right) \tag{1.1.8.3}$$

$$= -\frac{9}{2} + 20\left(\frac{3}{2}\right) = -\frac{9}{2} + 30 = \frac{51}{2} = 25\frac{1}{2} \tag{1.1.8.4}$$

1.1.9 For what value of k will the system of equations $2x + 3y = 7$ and $(k + 2)x - 3(1 - k)y = 5k + 1$ have infinitely many solutions? (10 2019)

Solution: Writing the system of linear equations in canonical matrix form $\mathbf{Ax} = \mathbf{b}$:

$$\begin{pmatrix} 2 & 3 \\ k+2 & 3k-3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 \\ 5k+1 \end{pmatrix} \tag{1.1.9.1}$$

The augmented matrix is

$$[\mathbf{A} \mid \mathbf{b}] = \begin{pmatrix} 2 & 3 & 7 \\ k+2 & 3k-3 & 5k+1 \end{pmatrix} \tag{1.1.9.2}$$

Performing the elementary row operation $R_2 \to 2R_2 - (k+2)R_1$:

$$[\mathbf{A} \mid \mathbf{b}] \xrightarrow{R_2 \to 2R_2-(k+2)R_1} \begin{pmatrix} 2 & 3 & 7 \\ 0 & 3k-12 & 3k-12 \end{pmatrix} \tag{1.1.9.3}$$

From (1.2.9.1), for the system to have infinitely many solutions:

$$\text{rank}\,(\mathbf{A}) = \text{rank}\,([\mathbf{A} \mid \mathbf{b}]) < 2 \tag{1.1.9.4}$$

This requires the second row to be entirely zero:

$$3k - 12 = 0 \implies k = 4 \tag{1.1.9.5}$$

1.1.10 Find the mode of the following frequency distribution:

Class Interval	25–30	30–35	35–40	40–45	45–50	50–55
Frequency	25	34	50	42	38	14

(10 2019)

Solution: The maximum frequency is 50, corresponding to the modal class 35–40. From the modal class:

$$l = 35, \quad h = 5, \quad f_1 = 50, \quad f_0 = 34, \quad f_2 = 42 \tag{1.1.10.1}$$

From (1.2.10.1), the mode is given by:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h \tag{1.1.10.2}$$

$$= 35 + \left(\frac{50 - 34}{2\,(50) - 34 - 42}\right) \times 5 \tag{1.1.10.3}$$

$$= 35 + \left(\frac{16}{24}\right) \times 5 = 35 + \frac{10}{3} \approx 38.33 \tag{1.1.10.4}$$

1.1.11 The larger of two supplementary angles exceeds the smaller by $18°$. Find the angles. (10 2019)

Solution: Let the larger angle be x and the smaller angle be y. Since the two angles are supplementary, their sum is $180°$:

$$x + y = 180° \tag{1.1.11.1}$$

The larger angle exceeds the smaller by $18°$:

$$x - y = 18° \tag{1.1.11.2}$$

Expressing the system of linear equations in canonical matrix form $\mathbf{Mx} = \mathbf{b}$:

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 180^\circ \\ 18^\circ \end{pmatrix} \tag{1.1.11.3}$$

From (1.2.15.1), evaluating the inverse matrix where $\left|\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\right| = -1 - 1 = -2$:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{-2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 180^\circ \\ 18^\circ \end{pmatrix} \tag{1.1.11.4}$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 180^\circ \\ 18^\circ \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 198^\circ \\ 162^\circ \end{pmatrix} = \begin{pmatrix} 99^\circ \\ 81^\circ \end{pmatrix} \tag{1.1.11.5}$$

Therefore, the two supplementary angles are 99° and 81°.

1.1.12 Sumit is 3 times as old as his son. Five years later, he shall be two and a half times as old as his son. How old is Sumit at present? (10 2019)

Solution: Let the present age of the son be x years and Sumit's present age be y years. From the given conditions:

$$y = 3x \implies 3x - y = 0 \tag{1.1.12.1}$$

Five years later, their ages will be $x + 5$ and $y + 5$ respectively:

$$y + 5 = \frac{5}{2}(x + 5) \implies 2y + 10 = 5x + 25 \implies 5x - 2y = -15 \tag{1.1.12.2}$$

Writing the system of equations in matrix form $\mathbf{Ax} = \mathbf{b}$:

$$\begin{pmatrix} 3 & -1 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ -15 \end{pmatrix} \tag{1.1.12.3}$$

From (1.2.15.1), since $\left|\mathbf{A}\right| = 3(-2) - (-1)(5) = -6 + 5 = -1$:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{-1} \begin{pmatrix} -2 & 1 \\ -5 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ -15 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 5 & -3 \end{pmatrix} \begin{pmatrix} 0 \\ -15 \end{pmatrix} = \begin{pmatrix} 15 \\ 45 \end{pmatrix} \tag{1.1.12.4}$$

Therefore, the present age of the son is 15 years and Sumit's present age is 45 years.

1.1.13 The probability of selecting a blue marble at random from a jar that contains only blue, black and green marbles is $\frac{1}{5}$. The probability of selecting a black marble at random from the same jar is $\frac{1}{4}$. If the jar contains 11 green marbles, find the total number of marbles in the jar. (10 2019)

Solution: Let the total number of marbles in the jar be n. Let B, K, and G denote the events of selecting a blue, black, and green marble, respectively. Since these events partition the sample space, from (1.2.11.1):

$$\Pr(B) + \Pr(K) + \Pr(G) = 1 \tag{1.1.13.1}$$

Substituting $\Pr(B) = \frac{1}{5}$ and $\Pr(K) = \frac{1}{4}$:

$$\frac{1}{5} + \frac{1}{4} + \Pr(G) = 1 \implies \frac{9}{20} + \Pr(G) = 1 \implies \Pr(G) = \frac{11}{20} \tag{1.1.13.2}$$

By the classical definition of probability:

$$\Pr(G) = \frac{n(G)}{n} \implies \frac{11}{20} = \frac{11}{n} \implies n = 20 \tag{1.1.13.3}$$

Therefore, the total number of marbles in the jar is 20.

1.1.14 Find a relation between x and y if the points $A(x, y)$, $B(-4, 6)$ and $C(-2, 3)$ are collinear. (10 2019)

Solution: Let the position vectors of the given points be

$$\mathbf{A} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} -2 \\ 3 \end{pmatrix} \tag{1.1.14.1}$$

Forming the displacement vectors relative to point $\mathbf{B}$:

$$\mathbf{v}_1 = \mathbf{A} - \mathbf{B} = \begin{pmatrix} x - (-4) \\ y - 6 \end{pmatrix} = \begin{pmatrix} x + 4 \\ y - 6 \end{pmatrix} \tag{1.1.14.2}$$

$$\mathbf{v}_2 = \mathbf{C} - \mathbf{B} = \begin{pmatrix} -2 - (-4) \\ 3 - 6 \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \end{pmatrix} \tag{1.1.14.3}$$

From (1.2.12.1), points $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are collinear if and only if the matrix formed by their displacement vectors has zero determinant:

$$\begin{vmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{vmatrix} = 0 \implies \begin{vmatrix} x + 4 & 2 \\ y - 6 & -3 \end{vmatrix} = 0 \tag{1.1.14.4}$$

Expanding the determinant:

$$-3(x + 4) - 2(y - 6) = 0 \implies -3x - 12 - 2y + 12 = 0 \implies 3x + 2y = 0 \tag{1.1.14.5}$$

The following code verifies the solution in Fig. 1.1.14.1:

codes/10/q14.py

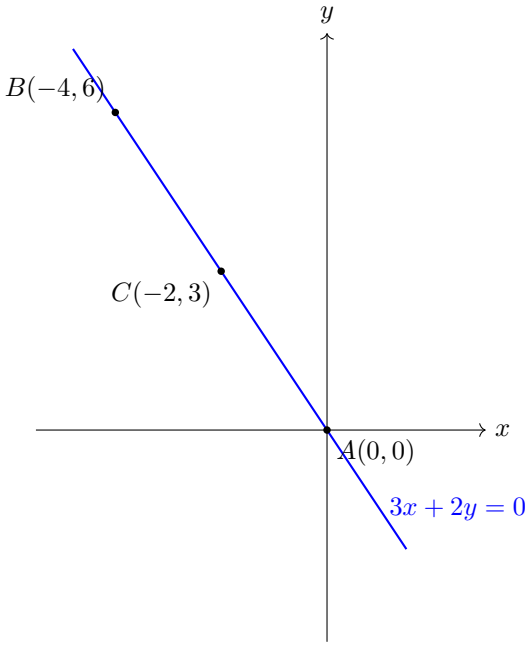


Fig. 1.1.14.1: Collinear points A, B, C on the line $3x + 2y = 0$

1.1.15 Find the area of a triangle whose vertices are $(1, -1)$, $(-4, 6)$ and $(-3, -5)$. (10 2019)

Solution: Let the position vectors of the vertices be

$$\mathbf{A} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} -3 \\ -5 \end{pmatrix} \tag{1.1.15.1}$$

Forming the two displacement edge vectors from reference vertex $\mathbf{A}$:

$$\mathbf{u} = \mathbf{B} - \mathbf{A} = \begin{pmatrix} -4-1 \\ 6-(-1) \end{pmatrix} = \begin{pmatrix} -5 \\ 7 \end{pmatrix} \tag{1.1.15.2}$$

$$\mathbf{v} = \mathbf{C} - \mathbf{A} = \begin{pmatrix} -3-1 \\ -5-(-1) \end{pmatrix} = \begin{pmatrix} -4 \\ -4 \end{pmatrix} \tag{1.1.15.3}$$

From (1.2.13.1), the area of $\triangle ABC$ is:

$$\text{Area} = \frac{1}{2} \left|\left| \mathbf{u} \quad \mathbf{v} \right|\right| = \frac{1}{2} \left\| \begin{matrix} -5 & -4 \\ 7 & -4 \end{matrix} \right\| \tag{1.1.15.4}$$

$$= \frac{1}{2} \left| (-5)(-4) - (-4)(7) \right| = \frac{1}{2} \left| 20 + 28 \right| = \frac{1}{2} (48) = 24 \text{ sq. units} \tag{1.1.15.5}$$

The following code verifies the solution in Fig. 1.1.15.1:

codes/10/q15.py

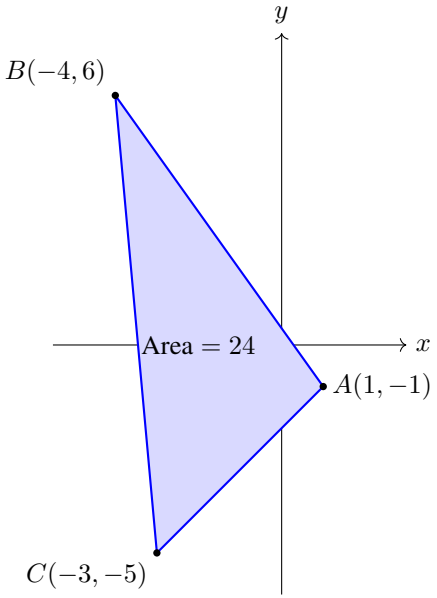


Fig. 1.1.15.1: Triangle ABC with vertices $A(1,-1)$, $B(-4,6)$, and $C(-3,-5)$

1.1.16 Write the smallest number which is divisible by both 306 and 657. (10 2019)

Solution: The smallest positive integer divisible by both numbers is their least common multiple, $\text{lcm}\,(306, 657)$. Applying Euclid's division lemma:

$$657 = 306 \times 2 + 45 \tag{1.1.16.1}$$
$$306 = 45 \times 6 + 36 \tag{1.1.16.2}$$
$$45 = 36 \times 1 + 9 \tag{1.1.16.3}$$
$$36 = 9 \times 4 + 0 \tag{1.1.16.4}$$

Since the remainder is 0, the greatest common divisor is:

$$\gcd\,(306, 657) = 9 \tag{1.1.16.5}$$

From (1.2.7.1):

$$\text{lcm}\,(306, 657) = \frac{306 \times 657}{\gcd\,(306, 657)} \tag{1.1.16.6}$$
$$= \frac{306 \times 657}{9} = 34 \times 657 = 22338 \tag{1.1.16.7}$$

1.1.17 Point A lies on the line segment XY joining $X(6,-6)$ and $Y(-4,-1)$ in such a way that $\frac{XA}{XY} = \frac{2}{5}$. If point A also lies on the line $3x + k(y+1) = 0$, find the value of k. (10 2019)

Solution: Point A lies on line segment XY with position vectors $\mathbf{X} = \begin{pmatrix} 6 \\ -6 \end{pmatrix}$ and

$\mathbf{Y} = \begin{pmatrix} -4 \\ -1 \end{pmatrix}$. Given $\frac{XA}{XY} = \frac{2}{5}$, since $XY = XA + AY$:

$$\frac{XA}{XA + AY} = \frac{2}{5} \implies 5XA = 2XA + 2AY \implies 3XA = 2AY \implies \frac{XA}{AY} = \frac{2}{3} \tag{1.1.17.1}$$

Thus, point $\mathbf{A}$ divides the line segment joining $\mathbf{X}$ and $\mathbf{Y}$ internally in the ratio $m : n = 2 : 3$. From (1.2.14.1):

$$\mathbf{A} = \frac{m\mathbf{Y} + n\mathbf{X}}{m + n} = \frac{2\begin{pmatrix} -4 \\ -1 \end{pmatrix} + 3\begin{pmatrix} 6 \\ -6 \end{pmatrix}}{2 + 3} \tag{1.1.17.2}$$

$$= \frac{1}{5}\left(\begin{pmatrix} -8 \\ -2 \end{pmatrix} + \begin{pmatrix} 18 \\ -18 \end{pmatrix}\right) = \frac{1}{5}\begin{pmatrix} 10 \\ -20 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \end{pmatrix} \tag{1.1.17.3}$$

Since point $\mathbf{A} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$ lies on the line $3x + k\,(y + 1) = 0$:

$$3\,(2) + k\,(-4 + 1) = 0 \implies 6 - 3k = 0 \implies 3k = 6 \implies k = 2 \tag{1.1.17.4}$$

The following code verifies the solution in Fig. 1.1.17.1:

codes/10/q17.py

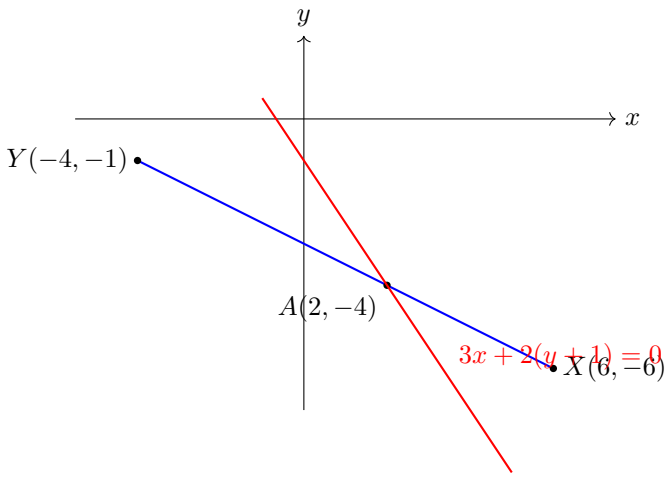


Fig. 1.1.17.1: Point $A(2, -4)$ dividing XY in ratio $2 : 3$

1.1.18 Solve for x:

$$x^2 + 5x - \left(a^2 + a - 6\right) = 0 \tag{1.1.18.1}$$

(10 2019)

Solution: Comparing with the standard quadratic equation $Ax^2 + Bx + C = 0$:

$$A = 1, \quad B = 5, \quad C = -\left(a^2 + a - 6\right) \tag{1.1.18.2}$$

From (1.2.6.1), the discriminant D is:

$$\begin{aligned} D = B^2 - 4AC &= 5^2 - 4\,(1)\,\left[-\left(a^2 + a - 6\right)\right] && (1.1.18.3)\\ &= 25 + 4\left(a^2 + a - 6\right) = 25 + 4a^2 + 4a - 24 && (1.1.18.4)\\ &= 4a^2 + 4a + 1 = (2a + 1)^2 && (1.1.18.5) \end{aligned}$$

Using the quadratic formula:

$$x = \frac{-B \pm \sqrt{D}}{2A} = \frac{-5 \pm \sqrt{(2a+1)^2}}{2\,(1)} = \frac{-5 \pm (2a+1)}{2} \tag{1.1.18.6}$$

Taking the positive sign:

$$x = \frac{-5 + (2a+1)}{2} = \frac{2a - 4}{2} = a - 2 \tag{1.1.18.7}$$

Taking the negative sign:

$$x = \frac{-5 - (2a+1)}{2} = \frac{-2a - 6}{2} = -\,(a + 3) \tag{1.1.18.8}$$

Therefore, the roots are $x = a - 2$ or $x = -\,(a + 3)$.

1.1.19 Find A and B if $\sin(A + 2B) = \frac{\sqrt{3}}{2}$ and $\cos(A + 4B) = 0$, where A and B are acute angles. (10 2019)

Solution: From (1.2.2.1), since $\sin(60°) = \frac{\sqrt{3}}{2}$ and angles are acute:

$$A + 2B = 60° \tag{1.1.19.1}$$

Since $\cos(90°) = 0$:

$$A + 4B = 90° \tag{1.1.19.2}$$

Expressing the system of linear equations in matrix form $\mathbf{Mx} = \mathbf{b}$:

$$\begin{pmatrix} 1 & 2 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 60° \\ 90° \end{pmatrix} \tag{1.1.19.3}$$

From (1.2.15.1), evaluating the determinant and inverse of $\mathbf{M}$:

$$\left|\mathbf{M}\right| = \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} = 1\,(4) - 2\,(1) = 2 \tag{1.1.19.4}$$

$$\mathbf{M}^{-1} = \frac{1}{2}\begin{pmatrix} 4 & -2 \\ -1 & 1 \end{pmatrix} \tag{1.1.19.5}$$

Solving for the angles:

$$\begin{pmatrix} A \\ B \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 4 & -2 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 60^\circ \\ 90^\circ \end{pmatrix} \tag{1.1.19.6}$$

$$= \frac{1}{2}\begin{pmatrix} 4\left(60^\circ\right) - 2\left(90^\circ\right) \\ -1\left(60^\circ\right) + 1\left(90^\circ\right) \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 240^\circ - 180^\circ \\ -60^\circ + 90^\circ \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 60^\circ \\ 30^\circ \end{pmatrix} = \begin{pmatrix} 30^\circ \\ 15^\circ \end{pmatrix} \tag{1.1.19.7}$$

Therefore, $A = 30^\circ$ and $B = 15^\circ$.

1.1.20 Prove that $2 + 5\sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is an irrational number. (10 2019)

Solution: Assume to the contrary that $2 + 5\sqrt{3}$ is a rational number. Then it can be written in the form $\frac{a}{b}$, where $a, b \in \mathbb{Z}$, $b \neq 0$, and $\gcd(a, b) = 1$:

$$2 + 5\sqrt{3} = \frac{a}{b} \tag{1.1.20.1}$$

Rearranging the terms:

$$5\sqrt{3} = \frac{a}{b} - 2 = \frac{a - 2b}{b} \tag{1.1.20.2}$$

$$\implies \sqrt{3} = \frac{a - 2b}{5b} \tag{1.1.20.3}$$

Since a and b are integers, $a - 2b$ and $5b$ are also integers with $5b \neq 0$. Thus, the right-hand side represents a rational number:

$$\frac{a - 2b}{5b} \in \mathbb{Q} \implies \sqrt{3} \in \mathbb{Q} \tag{1.1.20.4}$$

This contradicts the given fact that $\sqrt{3}$ is an irrational number. Hence, our assumption was false, and $2 + 5\sqrt{3}$ is an irrational number.

1.1.21 Using Euclid's Algorithm, find the HCF of 2048 and 960. (10 2019)

Solution: From (1.2.16.1), the reduction of an integer state vector under elementary unimodular shear matrices preserves the greatest common divisor:

$$\mathbf{v}_0 = \begin{pmatrix} 2048 \\ 960 \end{pmatrix} \tag{1.1.21.1}$$

Applying the first shear transformation $\mathbf{E}_1 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$:

$$\mathbf{v}_1 = \mathbf{E}_1\mathbf{v}_0 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2048 \\ 960 \end{pmatrix} = \begin{pmatrix} 2048 - 2\left(960\right) \\ 960 \end{pmatrix} = \begin{pmatrix} 128 \\ 960 \end{pmatrix} \tag{1.1.21.2}$$

Applying the second shear transformation $\mathbf{E}_2 = \begin{pmatrix} 1 & 0 \\ -7 & 1 \end{pmatrix}$:

$$\mathbf{v}_2 = \mathbf{E}_2\mathbf{v}_1 = \begin{pmatrix} 1 & 0 \\ -7 & 1 \end{pmatrix}\begin{pmatrix} 128 \\ 960 \end{pmatrix} = \begin{pmatrix} 128 \\ 960 - 7\left(128\right) \end{pmatrix} = \begin{pmatrix} 128 \\ 64 \end{pmatrix} \tag{1.1.21.3}$$

Applying the third shear transformation $\mathbf{E}_3 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$:

$$\mathbf{v}_3 = \mathbf{E}_3\mathbf{v}_2 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 128 \\ 64 \end{pmatrix} = \begin{pmatrix} 128 - 2\,(64) \\ 64 \end{pmatrix} = \begin{pmatrix} 0 \\ 64 \end{pmatrix} \tag{1.1.21.4}$$

The non-zero invariant entry of the reduced state vector yields:

$$\gcd\,(2048, 960) = 64 \tag{1.1.21.5}$$

The following code verifies the solution:

1.1.22 Two right triangles ABC and DBC are drawn on the same hypotenuse BC and on the same side of BC. If AC and BD intersect at P, prove that $AP \times PC = BP \times DP$. (10 2019)

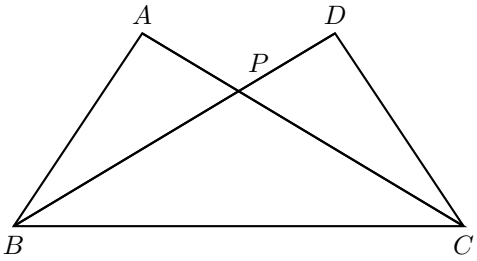


Fig. 1.1.22.1: Right triangles ABC and DBC on the common hypotenuse BC

Solution: In right triangles ABC and DBC, both are right-angled with hypotenuse BC:

$$\angle BAC = 90^\circ, \quad \angle BDC = 90^\circ \tag{1.1.22.1}$$

Since the hypotenuse BC subtends equal angles of 90° at vertices A and D on the same side of BC, points A, B, C, D are concyclic and lie on a circle with diameter BC. For the chord AD, angles in the same segment of the circle are equal:

$$\angle PBA = \angle PCD \tag{1.1.22.2}$$

In $\triangle PAB$ and $\triangle PDC$:

$$\angle APB = \angle DPC \quad \text{(vertically opposite angles)} \tag{1.1.22.3}$$
$$\angle PBA = \angle PCD \quad \text{(angles in the same segment)} \tag{1.1.22.4}$$

By the AA similarity criterion, $\triangle PAB \sim \triangle PDC$. From (1.2.17.1), the corresponding sides are proportional:

$$\frac{AP}{DP} = \frac{BP}{PC} \implies AP \times PC = BP \times DP \tag{1.1.22.5}$$

The following code verifies the geometric property:

1.1.23 Diagonals of a trapezium $PQRS$ intersect each other at the point O, $PQ \parallel RS$ and $PQ = 3RS$. Find the ratio of the areas of triangles POQ and ROS. (10 2019)

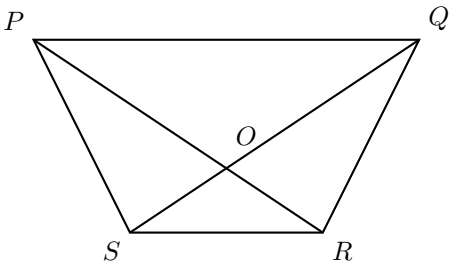


Fig. 1.1.23.1: Trapezium $PQRS$ with diagonals intersecting at O

Solution: In trapezium $PQRS$, given $PQ \parallel RS$. In $\triangle POQ$ and $\triangle ROS$:

$$\angle POQ = \angle ROS \quad \text{(vertically opposite angles)} \tag{1.1.23.1}$$
$$\angle OPQ = \angle ORS \quad \text{(alternate interior angles, since } PQ \parallel RS\text{)} \tag{1.1.23.2}$$
$$\angle OQP = \angle OSR \quad \text{(alternate interior angles, since } PQ \parallel RS\text{)} \tag{1.1.23.3}$$

Therefore, by the AAA similarity criterion, $\triangle POQ \sim \triangle ROS$. From (1.2.18.1):

$$\frac{\text{ar}\,(\triangle POQ)}{\text{ar}\,(\triangle ROS)} = \left(\frac{PQ}{RS}\right)^2 \tag{1.1.23.4}$$

Given that $PQ = 3RS$:

$$\frac{\text{ar}\,(\triangle POQ)}{\text{ar}\,(\triangle ROS)} = \left(\frac{3RS}{RS}\right)^2 = 3^2 = \frac{9}{1} \tag{1.1.23.5}$$

Thus, the ratio of the areas of $\triangle POQ$ and $\triangle ROS$ is $9 : 1$. The following code verifies the area ratio:

1.1.24 In Fig. 1.1.24.1, PQ and RS are two parallel tangents to a circle with centre O and another tangent AB with point of contact C intersects PQ at A and RS at B. Prove that $\angle AOB = 90°$. (10 2019)

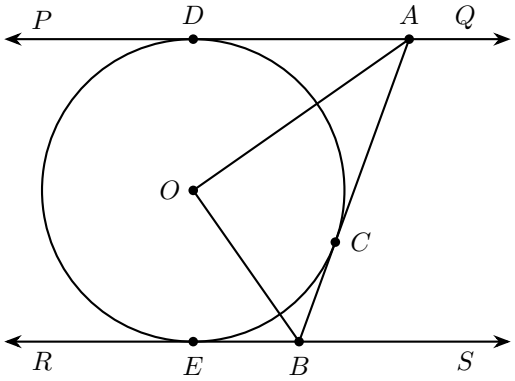


Fig. 1.1.24.1: Parallel tangents PQ, RS and transversal tangent AB

Solution: Let the centre O be the coordinate origin $\begin{pmatrix}0\\0\end{pmatrix}$. The parallel tangents PQ and RS correspond to the horizontal hyperplanes $y = r$ and $y = -r$. From (1.2.19.1), let the point of contact C on the circle $\|\mathbf{x}\| = r$ be parameterized by:

$$\mathbf{r}_C = \begin{pmatrix} r\cos\theta \\ r\sin\theta \end{pmatrix}, \quad \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \tag{1.1.24.1}$$

The tangent line at C with normal $\hat{n} = \begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix}$ has equation:

$$x\cos\theta + y\sin\theta = r \tag{1.1.24.2}$$

Intersecting with $y = r$ (line PQ) and $y = -r$ (line RS):

$$\mathbf{r}_A = \begin{pmatrix} r\frac{1-\sin\theta}{\cos\theta} \\ r \end{pmatrix}, \quad \mathbf{r}_B = \begin{pmatrix} r\frac{1+\sin\theta}{\cos\theta} \\ -r \end{pmatrix} \tag{1.1.24.3}$$

From (1.2.19.2), evaluating the scalar product $\mathbf{r}_A^\top \mathbf{r}_B$:

$$\mathbf{r}_A^\top \mathbf{r}_B = r^2 \left(\frac{1-\sin\theta}{\cos\theta}\right)\left(\frac{1+\sin\theta}{\cos\theta}\right) + r\,(-r) \tag{1.1.24.4}$$

$$= r^2\left(\frac{1-\sin^2\theta}{\cos^2\theta}\right) - r^2 = r^2\,(1) - r^2 = 0 \tag{1.1.24.5}$$

Since $\mathbf{r}_A^\top \mathbf{r}_B = 0$, $\mathbf{OA} \perp \mathbf{OB}$, which proves that $\angle AOB = 90°$. The following code verifies the orthogonality:

1.1.25 In Fig. 1.1.25.1, a square $OABC$ is inscribed in a quadrant $OPBQ$. If $OA = 15$ cm, find the area of the shaded region. (Use $\pi = 3.14$.) (10 2019)

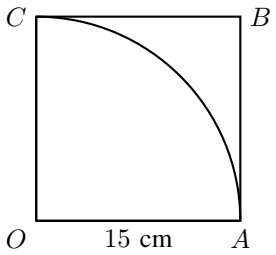


Fig. 1.1.25.1: Square $OABC$ inscribed in circular quadrant $OPBQ$

Solution: From (1.2.20.2), the side length of square $OABC$ is $s = OA = 15$ cm. Applying Pythagoras theorem (1.2.4.1) in right-angled $\triangle OAB$, diagonal OB forms the radius r of quadrant $OPBQ$:

$$r^2 = OB^2 = OA^2 + AB^2 = 15^2 + 15^2 = 450 \implies r = 15\sqrt{2} \text{ cm} \tag{1.1.25.1}$$

From (1.2.20.1), the area of quadrant $OPBQ$ is:

$$\text{Area of quadrant} = \frac{1}{4}\pi r^2 = \frac{1}{4}(3.14)(450) = \frac{1413}{4} = 353.25 \text{ cm}^2 \quad (1.1.25.2)$$

The area of square $OABC$ is:

$$\text{Area of square} = s^2 = 15^2 = 225 \text{ cm}^2 \quad (1.1.25.3)$$

Therefore, the area of the shaded region is:

$$\text{Area of shaded region} = \text{Area of quadrant} - \text{Area of square} \quad (1.1.25.4)$$

$$= 353.25 - 225 = 128.25 \text{ cm}^2 \quad (1.1.25.5)$$

The following code verifies the solution:

1.1.26 In Fig. 1.1.26.1, $ABCD$ is a square with side $2\sqrt{2}$ cm and inscribed in a circle. Find the area of the shaded region. (Use $\pi = 3.14$.) (10 2019)

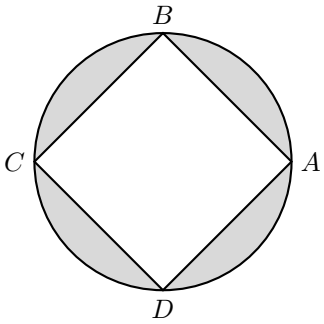


Fig. 1.1.26.1: Square $ABCD$ inscribed in circumscribed circle

Solution: Let the side of the inscribed square $ABCD$ be $s = 2\sqrt{2}$ cm. Since each interior angle of the square is $90°$, diagonal AC is the diameter of the circle. From (1.2.4.1):

$$AC^2 = AB^2 + BC^2 = \left(2\sqrt{2}\right)^2 + \left(2\sqrt{2}\right)^2 = 8 + 8 = 16 \implies AC = 4 \text{ cm} \quad (1.1.26.1)$$

Thus, the radius of the circle is:

$$r = \frac{AC}{2} = \frac{4}{2} = 2 \text{ cm} \quad (1.1.26.2)$$

From (1.2.20.3), the area of the circle is:

$$\text{Area of circle} = \pi r^2 = 3.14 \times 2^2 = 12.56 \text{ cm}^2 \quad (1.1.26.3)$$

The area of the inscribed square is:

$$\text{Area of square} = s^2 = \left(2\sqrt{2}\right)^2 = 8 \text{ cm}^2 \quad (1.1.26.4)$$

The area of the shaded region is:

$$\text{Area of shaded region} = \text{Area of circle} - \text{Area of square} \tag{1.1.26.5}$$

$$= 12.56 - 8 = 4.56 \text{ cm}^2 \tag{1.1.26.6}$$

The following code verifies the solution:

1.1.27 A solid is in the form of a cylinder with hemispherical ends. The total height of the solid is 20 cm and the diameter of the cylinder is 7 cm. Find the total volume of the solid. (Use $\pi = \frac{22}{7}$.) (10 2019)

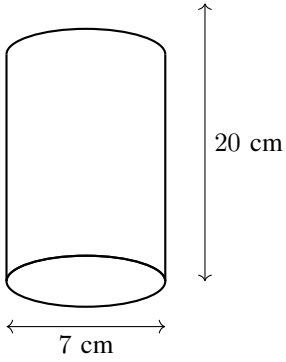


Fig. 1.1.27.1: Cylinder with hemispherical ends

Solution: The diameter is $d = 7$ cm, giving radius $r = \frac{7}{2}$ cm. Total height of the solid is $H = 20$ cm. The height of the cylindrical body h is:

$$h = H - 2r = 20 - 2\left(\frac{7}{2}\right) = 20 - 7 = 13 \text{ cm} \tag{1.1.27.1}$$

From (1.2.21.1), the volume of the cylindrical portion is:

$$V_{\text{cyl}} = \pi r^2 h = \frac{22}{7} \times \left(\frac{7}{2}\right)^2 \times 13 = \frac{22}{7} \times \frac{49}{4} \times 13 = \frac{1001}{2} = 500.5 \text{ cm}^3 \tag{1.1.27.2}$$

From (1.2.21.2), the two hemispherical ends combine to form a sphere:

$$V_{\text{ends}} = 2 \times \left(\frac{2}{3}\pi r^3\right) = \frac{4}{3}\pi r^3 = \frac{4}{3} \times \frac{22}{7} \times \left(\frac{7}{2}\right)^3 = \frac{539}{3} \approx 179.67 \text{ cm}^3 \tag{1.1.27.3}$$

The total volume of the solid is:

$$V = V_{\text{cyl}} + V_{\text{ends}} = \frac{1001}{2} + \frac{539}{3} = \frac{3003 + 1078}{6} = \frac{4081}{6} \approx 680.17 \text{ cm}^3 \tag{1.1.27.4}$$

The following code verifies the computed volume:

codes/10/q27.py

1.1.28 The marks obtained by 100 students in an examination are given below:

Marks	30–35	35–40	40–45	45–50	50–55	55–60	60–65
Number of Students	14	16	28	23	18	8	3

Find the mean marks of the students. (10 2019)

Solution: Constructing the computation table with class marks $x_i = \frac{l_i+u_i}{2}$:

Class Interval	Class Mark (x_i)	Frequency (f_i)	$f_i x_i$
30–35	32.5	14	455.0
35–40	37.5	16	600.0
40–45	42.5	28	1190.0
45–50	47.5	23	1092.5
50–55	52.5	18	945.0
55–60	57.5	8	460.0
60–65	62.5	3	187.5
Total		$\sum f_i = 110$	$\sum f_i x_i = 4930.0$

From (1.2.22.1), the arithmetic mean is:

$$\bar{x} = \frac{\sum_{i=1}^{k} f_i x_i}{\sum_{i=1}^{k} f_i} = \frac{4930}{110} = \frac{493}{11} \approx 44.82 \tag{1.1.28.1}$$

The following code verifies the solution in Fig. **??**:

codes/10/q28.py

1.1.29 For what value of k is the polynomial $f(x) = 3x^4 - 9x^3 + x^2 + 15x + k$ completely divisible by $3x^2 - 5$? (10 2019)

Solution: From (1.2.23.1), work in the quotient ring $\mathbb{R}[x]/\langle 3x^2 - 5\rangle$, where the divisor induces the reduction relation:

$$3x^2 - 5 \equiv 0 \implies x^2 \equiv \frac{5}{3} \tag{1.1.29.1}$$

Higher powers of the state variable x reduce algebraically:

$$x^3 = x \cdot x^2 \equiv \frac{5}{3}x \tag{1.1.29.2}$$

$$x^4 = \left(x^2\right)^2 \equiv \left(\frac{5}{3}\right)^2 = \frac{25}{9} \tag{1.1.29.3}$$

Projecting $f(x)$ modulo $3x^2 - 5$:

$$f(x) \equiv 3\left(\frac{25}{9}\right) - 9\left(\frac{5}{3}x\right) + \left(\frac{5}{3}\right) + 15x + k \tag{1.1.29.4}$$

$$\equiv \frac{25}{3} - 15x + \frac{5}{3} + 15x + k \tag{1.1.29.5}$$

$$\equiv \frac{30}{3} + k \equiv 10 + k \pmod{3x^2 - 5} \tag{1.1.29.6}$$

For $f(x)$ to be completely divisible, the remainder residue must be zero:

$$10 + k = 0 \implies k = -10 \tag{1.1.29.7}$$

The following code verifies the polynomial division:

codes/10/q29.py

1.1.30 Find the zeroes of the quadratic polynomial $7y^2 - \frac{11}{3}y - \frac{2}{3}$ and verify the relationship between the zeroes and the coefficients. (10 2019)

Solution: To find the zeroes, set $p(y) = 7y^2 - \frac{11}{3}y - \frac{2}{3} = 0$. Multiplying by 3:

$$21y^2 - 11y - 2 = 0 \tag{1.1.30.1}$$

Factoring by splitting the middle term:

$$21y^2 - 14y + 3y - 2 = 0 \tag{1.1.30.2}$$

$$7y\,(3y - 2) + 1\,(3y - 2) = 0 \tag{1.1.30.3}$$

$$(3y - 2)\,(7y + 1) = 0 \tag{1.1.30.4}$$

Equating each factor to zero:

$$3y - 2 = 0 \implies y = \frac{2}{3}, \qquad 7y + 1 = 0 \implies y = -\frac{1}{7} \tag{1.1.30.5}$$

Thus, the zeroes are $\alpha = \frac{2}{3}$ and $\beta = -\frac{1}{7}$.
Comparing $p(y)$ with the standard quadratic polynomial $ay^2 + by + c$:

$$a = 7, \quad b = -\frac{11}{3}, \quad c = -\frac{2}{3} \tag{1.1.30.6}$$

Verifying Vieta's relations (1.2.24.1) and (1.2.24.2):

$$\alpha + \beta = \frac{2}{3} + \left(-\frac{1}{7}\right) = \frac{14 - 3}{21} = \frac{11}{21} = -\frac{-11/3}{7} = -\frac{b}{a} \tag{1.1.30.7}$$

$$\alpha\beta = \left(\frac{2}{3}\right)\left(-\frac{1}{7}\right) = -\frac{2}{21} = \frac{-2/3}{7} = \frac{c}{a} \tag{1.1.30.8}$$

Hence, the relationship between zeroes and coefficients is verified.

1.1.31 Prove that the ratio of the areas of two similar triangles is equal to the ratio of the squares on their corresponding sides. (10 2019)

Solution: Let $\triangle ABC \sim \triangle PQR$. Draw altitudes $AM \perp BC$ and $PN \perp QR$. The ratio of their areas is:

$$\frac{\text{ar}\,(\triangle ABC)}{\text{ar}\,(\triangle PQR)} = \frac{\frac{1}{2}BC \times AM}{\frac{1}{2}QR \times PN} = \left(\frac{BC}{QR}\right)\left(\frac{AM}{PN}\right) \tag{1.1.31.1}$$

In $\triangle ABM$ and $\triangle PQN$, $\angle B = \angle Q$ (as $\triangle ABC \sim \triangle PQR$) and $\angle AMB = \angle PNQ = 90^\circ$. By AA similarity:

$$\triangle ABM \sim \triangle PQN \implies \frac{AM}{PN} = \frac{AB}{PQ} \tag{1.1.31.2}$$

Since $\triangle ABC \sim \triangle PQR$, the corresponding sides are proportional:

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP} \tag{1.1.31.3}$$

Substituting $\frac{AM}{PN} = \frac{BC}{QR}$ yields (1.2.18.1):

$$\frac{\text{ar}\,(\triangle ABC)}{\text{ar}\,(\triangle PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{BC}{QR}\right)^2 = \left(\frac{CA}{FD}\right)^2 \tag{1.1.31.4}$$

1.1.32 Two poles of equal heights are standing opposite to each other on either side of the road which is 80 m wide. From a point P between them on the road, the angle of elevation of the top of a pole is 60° and the angle of depression from the top of the other pole to point P is 30°. Find the heights of the poles and the distance of the point P from the poles. (10 2019)

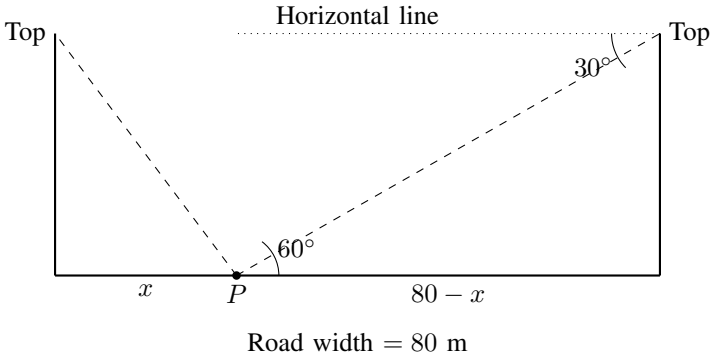


Fig. 1.1.32.1: Poles and point of observation on road

Solution: Let the height of both poles be h and the distance of P from the first pole be x. From the angles of elevation and depression:

$$\tan(60^\circ) = \frac{h}{x} \implies h = \sqrt{3}x \tag{1.1.32.1}$$

$$\tan(30^\circ) = \frac{h}{80-x} \implies h = \frac{80-x}{\sqrt{3}} \tag{1.1.32.2}$$

Equating (1.1.32.1) and (1.1.32.2):

$$\sqrt{3}x = \frac{80-x}{\sqrt{3}} \implies 3x = 80 - x \implies 4x = 80 \implies x = 20 \text{ m} \tag{1.1.32.3}$$

The height of each pole is:

$$h = 20\sqrt{3} \text{ m} \tag{1.1.32.4}$$

The distance from the second pole is $80 - 20 = 60$ m.

1.1.33 The total cost of a certain length of a piece of cloth is Rs. 200. If the piece was 5 m longer and each metre of cloth costs Rs. 2 less, the cost of the piece would have

remained unchanged. How long is the piece and what is its original rate per metre? (10 2019)

Solution: Let the original length of cloth be x metres. The original rate per metre is $\frac{200}{x}$. With the new conditions:

$$(x+5)\left(\frac{200}{x}-2\right)=200 \tag{1.1.33.1}$$

Expanding:

$$200-2x+\frac{1000}{x}-10=200 \tag{1.1.33.2}$$

$$-2x-10+\frac{1000}{x}=0 \tag{1.1.33.3}$$

$$x^2+5x-500=0 \tag{1.1.33.4}$$

Factoring:

$$(x+25)(x-20)=0 \implies x=20 \text{ m} \quad (\text{since } x>0) \tag{1.1.33.5}$$

The original rate is:

$$\text{Rate}=\frac{200}{20}=10 \text{ Rs./m} \tag{1.1.33.6}$$

1.1.34 Which term of the Arithmetic Progression -7, -12, -17, -22, ... will be -82? Is -100 any term of the A.P.? Give reason for your answer. (10 2019)

Solution: Here, $a=-7$ and $d=-12-(-7)=-5$. From (**??**):

$$a_n=a+(n-1)\,d \tag{1.1.34.1}$$

Setting $a_n=-82$:

$$-82=-7+(n-1)(-5) \implies -75=-5(n-1) \implies n=16 \tag{1.1.34.2}$$

Thus, the 16th term is -82. Now testing -100:

$$-100=-7+(m-1)(-5) \implies -93=-5(m-1) \implies m=1+\frac{93}{5}=19.6 \tag{1.1.34.3}$$

Since $m \notin \mathbb{Z}k$, -100 is not a term of the A.P. The following code verifies the solution:

1.1.35 Change the following data into a "less than type" distribution and draw its ogive:

Class Interval	30–40	40–50	50–60	60–70	70–80	80–90	90–100
Frequency	7	5	8	10	6	6	8

(10 2019)

Solution: The cumulative frequency table for the "less than type" distribution using the upper class limits:

Marks (Upper Limit)	Frequency (f)	Cumulative Frequency (cf)
Less than 40	7	7
Less than 50	5	12
Less than 60	8	20
Less than 70	10	30
Less than 80	6	36
Less than 90	6	42
Less than 100	8	50
Total	$N = 50$	

Plotting points $(30, 0)$, $(40, 7)$, $(50, 12)$, $(60, 20)$, $(70, 30)$, $(80, 36)$, $(90, 42)$, $(100, 50)$ on the Cartesian plane yields the ogive curve. Here, total observations $N = 50$:

$$\frac{N}{2} = \frac{50}{2} = 25 \tag{1.1.35.1}$$

Locating 25 on the y-axis and projecting onto the ogive curve, the corresponding value on the x-axis gives the median:

$$\text{Median} = 65 \tag{1.1.35.2}$$

1.1.36 Draw two concentric circles of radii 2 cm and 5 cm. Take a point P on the outer circle and construct a pair of tangents PA and PB to the smaller circle. Measure PA. (10 2019)

Solution: Let the common centre be $\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. Point $\mathbf{P}$ lies on the outer circle such that $\|\mathbf{P}\| = 5$ cm. Since PA is tangent to the inner circle of radius $r = 2$ cm at point A, radius $\mathbf{OA}$ is orthogonal to tangent $\mathbf{PA}$ ($\mathbf{OA} \perp \mathbf{PA}$). Applying Pythagoras theorem (1.2.4.1) in right-angled $\triangle OAP$:

$$PA^2 = OP^2 - OA^2 = 5^2 - 2^2 = 25 - 4 = 21 \implies PA = \sqrt{21} \approx 4.58 \text{ cm} \tag{1.1.36.1}$$

The following code generates the geometric construction:

codes/10/q36.py

1.1.37 A solid iron pole consists of a cylinder of height 220 cm and base diameter 24 cm, which is surmounted by another cylinder of height 60 cm and radius 8 cm. Find the mass of the pole, given that 1 cm^3 of iron has approximately 8 gm mass. (Use $\pi = 3.14$.) (10 2019)

Solution: The iron pole consists of two coaxial right circular cylinders stacked vertically.

Lower (Larger) Cylinder:

$$h_1 = 220 \text{ cm}, \quad r_1 = \frac{24}{2} = 12 \text{ cm} \tag{1.1.37.1}$$

From (1.2.21.1), the volume of the lower cylinder is:

$$V_1 = \pi r_1^2 h_1 = 3.14 \times 12^2 \times 220 = 3.14 \times 144 \times 220 = 99475.2 \text{ cm}^3 \tag{1.1.37.2}$$

Upper (Smaller) Cylinder:

$$h_2 = 60 \text{ cm}, \quad r_2 = 8 \text{ cm} \tag{1.1.37.3}$$

The volume of the upper cylinder is:

$$V_2 = \pi r_2^2 h_2 = 3.14 \times 8^2 \times 60 = 3.1464 \times 60 = 12057.6 \text{ cm}^3 \tag{1.1.37.4}$$

Total Volume:

$$V = V_1 + V_2 = 99475.2 + 12057.6 = 111532.8 \text{ cm}^3 \tag{1.1.37.5}$$

Total Mass: From (1.2.21.3), with density $\rho = 8$ gm/cm^3:

$$\text{Mass} = V \times \rho = 111532.8 \times 8 = 892262.4 \text{ gm} = 892.26 \text{ kg} \tag{1.1.37.6}$$

5

B. Formula-10

1.2.1 Distance Formula and Equidistant Locus. The Euclidean distance between two points $\mathbf{x}_1 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ and $\mathbf{x}_2 = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ in $\mathbb{R}^2$ is given by

$$\|\mathbf{x}_1 - \mathbf{x}_2\|^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2 \tag{1.2.1.1}$$

A point $\mathbf{x}$ is equidistant from two fixed points $\mathbf{a}$ and $\mathbf{b}$ if

$$\|\mathbf{x} - \mathbf{a}\|^2 = \|\mathbf{x} - \mathbf{b}\|^2 \tag{1.2.1.2}$$

Solution: Expanding (1.2.1.2):

$$\|\mathbf{x}\|^2 - 2\mathbf{a}^\top \mathbf{x} + \|\mathbf{a}\|^2 = \|\mathbf{x}\|^2 - 2\mathbf{b}^\top \mathbf{x} + \|\mathbf{b}\|^2 \tag{1.2.1.3}$$

$$\implies 2(\mathbf{b} - \mathbf{a})^\top \mathbf{x} = \|\mathbf{b}\|^2 - \|\mathbf{a}\|^2 \tag{1.2.1.4}$$

yielding the equation of the perpendicular bisector.

1.2.2 Standard Trigonometric Values and Complementary Identity:

$$\sin(60^\circ) = \frac{\sqrt{3}}{2}, \quad \tan(45^\circ) = 1, \quad \cos(30^\circ) = \frac{\sqrt{3}}{2} \tag{1.2.2.1}$$

For complementary angles:

$$\cos(90^\circ - \theta) = \sin\theta \implies \cos(30^\circ) = \sin(60^\circ) \tag{1.2.2.2}$$

1.2.3 Fundamental Pythagorean Trigonometric Identities:

$$\cos^2 A + \sin^2 A = 1 \tag{1.2.3.1}$$

Solution: From (1.2.3.1), solving for $\cos A$ and its reciprocal $\sec A$:

$$\cos A = \sqrt{1 - \sin^2 A} \tag{1.2.3.2}$$

$$\sec A = \frac{1}{\cos A} = \frac{1}{\sqrt{1 - \sin^2 A}} \tag{1.2.3.3}$$

1.2.4 Pythagoras Theorem. In a right-angled triangle with legs a, b and hypotenuse c:

$$c^2 = a^2 + b^2 \tag{1.2.4.1}$$

For an isosceles right triangle where $a = b$:

$$c = \sqrt{a^2 + a^2} = a\sqrt{2} \tag{1.2.4.2}$$

1.2.5 Basic Proportionality Theorem (Thales' Theorem). If a line is drawn parallel to one side of a triangle intersecting the other two sides, then it divides the two sides in the same ratio:

$$\frac{AD}{BD} = \frac{AE}{CE} \tag{1.2.5.1}$$

Solution: Let h_1 be the altitude from E to AB and h_2 be the altitude from D to AC.

$$\frac{\text{ar}\,(\triangle ADE)}{\text{ar}\,(\triangle BDE)} = \frac{\frac{1}{2}AD \cdot h_1}{\frac{1}{2}BD \cdot h_1} = \frac{AD}{BD} \tag{1.2.5.2}$$

$$\frac{\text{ar}\,(\triangle ADE)}{\text{ar}\,(\triangle CDE)} = \frac{\frac{1}{2}AE \cdot h_2}{\frac{1}{2}CE \cdot h_2} = \frac{AE}{CE} \tag{1.2.5.3}$$

Since $\triangle BDE$ and $\triangle CDE$ share base DE between parallel lines $DE \parallel BC$, $\text{ar}\,(\triangle BDE) = \text{ar}\,(\triangle CDE)$, yielding (1.2.5.1).

1.2.6 Discriminant and Nature of Quadratic Roots. For the general quadratic equation $ax^2 + bx + c = 0$ ($a \neq 0$), the discriminant D is defined as

$$D = b^2 - 4ac \tag{1.2.6.1}$$

- $D > 0$: Two distinct real roots.
- $D = 0$: Two equal real roots.
- $D < 0$: No real roots (complex conjugate pair).

1.2.7 Product and GCD/LCM Duality for Two Positive Integers:

$$\gcd(a, b) \times \text{lcm}\,(a, b) = a \times b \tag{1.2.7.1}$$

Solution: Let $a = \prod p_i^{\alpha_i}$ and $b = \prod p_i^{\beta_i}$. Then:

$$\gcd(a, b) = \prod p_i^{\min(\alpha_i, \beta_i)}, \quad \text{lcm}\,(a, b) = \prod p_i^{\max(\alpha_i, \beta_i)} \tag{1.2.7.2}$$

Since $\min(\alpha_i, \beta_i) + \max(\alpha_i, \beta_i) = \alpha_i + \beta_i$, multiplying yields (1.2.7.1).

1.2.8 Sum of the First n Terms of an Arithmetic Progression:

$$S_n = \frac{n}{2}\left[2a + (n-1)\,d\right] \tag{1.2.8.1}$$

Solution: Writing S_n forward and backward:

$$S_n = a + (a+d) + \cdots + (a + (n-1)d) \tag{1.2.8.2}$$

$$S_n = (a + (n-1)d) + \cdots + (a+d) + a \tag{1.2.8.3}$$

Summing the two equations term-by-term:

$$2S_n = n\left[2a + (n-1)\,d\right] \tag{1.2.8.4}$$

yielding (1.2.8.1).

1.2.9 Consistency and Infinite Solutions of Linear Systems. For the matrix system $\mathbf{Ax} = \mathbf{b}$, infinite solutions exist if and only if

$$\text{rank}\,(\mathbf{A}) = \text{rank}\,([\mathbf{A} \mid \mathbf{b}]) < n \tag{1.2.9.1}$$

where n is the number of unknown variables.

1.2.10 Mode of Grouped Frequency Distribution:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h \tag{1.2.10.1}$$

where:

- l is the lower limit of the modal class
- h is the class interval width
- f_1 is the frequency of the modal class
- f_0 is the frequency of the preceding class
- f_2 is the frequency of the succeeding class

1.2.11 Classical Probability and Partition Axiom. If a sample space S is partitioned into pairwise disjoint events $E_1, E_2, \ldots, E_k$:

$$\sum_{i=1}^{k} \Pr\left(E_i\right) = 1, \quad \text{where } \Pr\left(E\right) = \frac{n(E)}{n(S)} \tag{1.2.11.1}$$

1.2.12 Collinearity of Three Points in $\mathbb{R}^2$. Points $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are collinear if and only if their displacement vectors are linearly dependent:

$$\left|\mathbf{B} - \mathbf{A} \quad \mathbf{C} - \mathbf{A}\right| = 0 \tag{1.2.12.1}$$

1.2.13 Area of a Triangle using Vector Determinant:

$$\text{Area}\left(\triangle ABC\right) = \frac{1}{2}\left|\left|\mathbf{B} - \mathbf{A} \quad \mathbf{C} - \mathbf{A}\right|\right| \tag{1.2.13.1}$$

1.2.14 Section Formula in Vector Notation. The point $\mathbf{P}$ dividing the segment between $\mathbf{A}$ and $\mathbf{B}$ internally in ratio $m : n$ is given by

$$\mathbf{P} = \frac{m\mathbf{B} + n\mathbf{A}}{m+n} \tag{1.2.14.1}$$

1.2.15 Matrix Inverse and Cramer's Rule for 2×2 Systems. If $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $\left|\mathbf{M}\right| = ad - bc \neq 0$:

$$\mathbf{M}^{-1} = \frac{1}{\left|\mathbf{M}\right|}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \tag{1.2.15.1}$$

The unique solution to $\mathbf{Mx} = \mathbf{b}$ is $\mathbf{x} = \mathbf{M}^{-1}\mathbf{b}$.

1.2.16 Euclidean Algorithm via Elementary Shear Transformations. The reduction of an integer vector $\mathbf{v}_0 = \begin{pmatrix} a \\ b \end{pmatrix}$ using shear matrices $\mathbf{E}_k = \begin{pmatrix} 1 & -q_k \\ 0 & 1 \end{pmatrix}$ or $\begin{pmatrix} 1 & 0 \\ -q_k & 1 \end{pmatrix}$ satisfies

$$\left|\mathbf{E}_k\right| = 1 \implies \gcd(a, b) = \gcd(\mathbf{E}_k \mathbf{v}_0) \tag{1.2.16.1}$$

terminating when one coordinate becomes zero, leaving the non-zero invariant as $\gcd(a, b)$.

1.2.17 Intersecting Chords Theorem. If two chords AC and BD of a circle intersect at an interior point P:

$$AP \times PC = BP \times DP \tag{1.2.17.1}$$

Solution: In $\triangle PAB$ and $\triangle PDC$, angles in the same circular segment are equal ($\angle PBA = \angle PCD$) and vertically opposite angles are equal ($\angle APB = \angle DPC$). By AA similarity:

$$\triangle PAB \sim \triangle PDC \implies \frac{AP}{DP} = \frac{BP}{PC} \tag{1.2.17.2}$$

Cross-multiplying yields (1.2.17.1).

1.2.18 Ratio of Areas of Similar Triangles. If $\triangle ABC \sim \triangle DEF$, then

$$\frac{\text{ar}\,(\triangle ABC)}{\text{ar}\,(\triangle DEF)} = \left(\frac{AB}{DE}\right)^2 = \left(\frac{BC}{EF}\right)^2 = \left(\frac{CA}{FD}\right)^2 \tag{1.2.18.1}$$

1.2.19 Tangent Hyperplane to a Circle. The tangent line to the circle $\|\mathbf{x}\| = r$ at contact point $\mathbf{r}_C = \begin{pmatrix} r\cos\theta \\ r\sin\theta \end{pmatrix}$ has normal vector $\hat{n} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$ and equation

$$\hat{n}^\top \mathbf{x} = r \iff x\cos\theta + y\sin\theta = r \tag{1.2.19.1}$$

Two vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if

$$\mathbf{u}^\top \mathbf{v} = 0 \iff \mathbf{u} \perp \mathbf{v} \tag{1.2.19.2}$$

1.2.20 Mensuration of Standard Plane and Composite Figures:

$$\text{Area of quadrant} = \frac{1}{4}\pi r^2 \tag{1.2.20.1}$$

$$\text{Area of square} = s^2, \quad \text{Diagonal } d = s\sqrt{2} \tag{1.2.20.2}$$

$$\text{Area of circumscribed circle} = \pi R^2 = \pi\left(\frac{s\sqrt{2}}{2}\right)^2 = \frac{\pi s^2}{2} \tag{1.2.20.3}$$

1.2.21 Mensuration of Cylindrical and Spherical Solids:

$$V_{\text{cylinder}} = \pi r^2 h \tag{1.2.21.1}$$

$$V_{\text{hemisphere}} = \frac{2}{3}\pi r^3 \implies V_{\text{sphere}} = \frac{4}{3}\pi r^3 \tag{1.2.21.2}$$

$$\text{Mass} = \rho \times V \tag{1.2.21.3}$$

where ρ is the mass density and V is the total volume.

1.2.22 Arithmetic Mean of Grouped Data (Direct Method):

$$\bar{x} = \frac{\sum_{i=1}^{k} f_i x_i}{\sum_{i=1}^{k} f_i}, \quad \text{where } x_i = \frac{l_i + u_i}{2} \tag{1.2.22.1}$$

1.2.23 Polynomial Division and Quotient Ring Reduction. In the polynomial ring $\mathbb{R}[x]$ modulo ideal $\langle d(x) \rangle$:

$$f(x) \equiv r(x) \pmod{d(x)}, \quad \deg(r) < \deg(d) \tag{1.2.23.1}$$

If $d(x)$ divides $f(x)$, then $r(x) \equiv 0$.

1.2.24 Vieta's Relations for a Quadratic Polynomial. If α and β are zeroes of $p(x) = ax^2 + bx + c$:

$$\alpha + \beta = -\frac{b}{a} \tag{1.2.24.1}$$

$$\alpha\beta = \frac{c}{a} \tag{1.2.24.2}$$

1.2.25 Algebraic Identities for Trigonometric Proofs:

$$a^3 - b^3 = (a - b)\left(a^2 + ab + b^2\right) \tag{1.2.25.1}$$

$$\csc^2\theta - \cot^2\theta = 1 \tag{1.2.25.2}$$

$$\sec^2\theta - \tan^2\theta = 1 \tag{1.2.25.3}$$

C. Class-12

1.3.1 Find $\left|\mathbf{AB}\right|$, if

$$\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 0 & 2 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 3 & 5 \\ 0 & 0 \end{pmatrix} \tag{1.3.1.1}$$

(12 2019)

Solution: From (1.4.1.1):

$$\left|\mathbf{AB}\right| = \left|\mathbf{A}\right| \left|\mathbf{B}\right| \tag{1.3.1.2}$$

From (1.4.1.2), evaluating the individual determinants:

$$\left|\mathbf{A}\right| = \left|\begin{pmatrix} 0 & -1 \\ 0 & 2 \end{pmatrix}\right| = (0)(2) - (-1)(0) = 0 \tag{1.3.1.3}$$

$$\left|\mathbf{B}\right| = \left|\begin{pmatrix} 3 & 5 \\ 0 & 0 \end{pmatrix}\right| = (3)(0) - (5)(0) = 0 \tag{1.3.1.4}$$

Therefore:

$$\left|\mathbf{AB}\right| = 0 \times 0 = 0 \tag{1.3.1.5}$$

1.3.2 Differentiate e^{3x} with respect to x. (12 2019)

Solution: Let the function be $y = e^{3x}$. From (1.4.2.2), differentiating with respect to x using the chain rule (1.4.2.1):

$$\frac{dy}{dx} = \frac{d}{dx}\left[e^{3x}\right] = 3e^{3x} \tag{1.3.2.1}$$

1.3.3 Find the order and degree (if defined) of the differential equation

$$\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^2 = 2x^2 \ln\left(\frac{d^2y}{dx^2}\right) \tag{1.3.3.1}$$

(12 2019)

Solution: Following the criteria in (1.4.4.1):

- The highest-order derivative appearing in (1.3.3.1) is $\frac{d^2y}{dx^2}$. Hence, Order $= 2$.
- The highest-order derivative $\frac{d^2y}{dx^2}$ appears inside a transcendental logarithmic function $\ln(\cdot)$, meaning the differential equation cannot be expressed as a polynomial in its derivatives. Therefore, the **Degree is not defined**.

1.3.4 Find the direction cosines of a line which makes equal angles with the coordinate axes. (12 2019)

Solution: Let the line make equal angles α with the coordinate axes. From (1.4.4.1) and (1.4.4.2):

$$l^2 + m^2 + n^2 = 1 \implies 3l^2 = 1 \implies l = \pm\frac{1}{\sqrt{3}} \tag{1.3.4.1}$$

From (1.4.4.3), since $l = m = n$, the direction cosines are:

$$\left(\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}\right) \tag{1.3.4.2}$$

1.3.5 A line passes through the point with position vector $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$ and is in the direction of the vector $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$. Find the equation of the line in Cartesian form. (12 2019)

Solution: From (1.4.5.2), with point $\mathbf{a} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$ and direction vector $\mathbf{b} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$:

$$\frac{x-2}{1} = \frac{y-(-1)}{1} = \frac{z-4}{-2} \tag{1.3.5.1}$$

Simplifying:

$$\frac{x-2}{1} = \frac{y+1}{1} = \frac{z-4}{-2} \tag{1.3.5.2}$$

The following code gives the plot

```
codes/12/q5.py
```

1.3.6 Find $\int \left(3 - 2x - x^2\right)\, dx$. (12 2019)

Solution: Let $I = \int \left(3 - 2x - x^2\right)\, dx$. Applying linearity and the power integration rule (1.4.6.1):

$$I = \int 3\, dx - \int 2x\, dx - \int x^2\, dx \tag{1.3.6.1}$$

$$= 3x - 2\left(\frac{x^2}{2}\right) - \frac{x^3}{3} + C \tag{1.3.6.2}$$

$$= 3x - x^2 - \frac{x^3}{3} + C \tag{1.3.6.3}$$

1.3.7 If $\mathbf{A} = \begin{pmatrix} p & 2 \\ 2 & p \end{pmatrix}$ and $\left|\mathbf{A}^3\right| = 125$, find the values of p. (12 2019)

Solution: From (1.4.1.2), evaluating the determinant of $\mathbf{A}$:

$$\left|\mathbf{A}\right| = \left|\begin{pmatrix} p & 2 \\ 2 & p \end{pmatrix}\right| = p^2 - 4 \tag{1.3.7.1}$$

From (1.4.7.1):

$$\left|\mathbf{A}^3\right| = \left(\left|\mathbf{A}\right|\right)^3 = \left(p^2 - 4\right)^3 \tag{1.3.7.2}$$

Given that $\left|\mathbf{A}^3\right| = 125$:

$$\left(p^2 - 4\right)^3 = 5^3 \implies p^2 - 4 = 5 \implies p^2 = 9 \implies p = \pm 3 \tag{1.3.7.3}$$

Thus, the values of p are 3 and -3.

1.3.8 Examine whether the operation $*$ defined on the set of all real numbers by $a * b = a^2 + b^2$ is a binary operation or not, and if it is a binary operation, find whether it is associative or not. (12 2019)

Solution: From (1.4.8.1), for any $a, b \in \mathbb{R}$, $a^2 \in \mathbb{R}$ and $b^2 \in \mathbb{R}$. Since the sum of real numbers is real, $a^2 + b^2 \in \mathbb{R}$. Hence, $*$ is a well-defined binary operation on $\mathbb{R}$. To test associativity from (1.4.8.2):

$$\text{LHS:} \quad (a * b) * c = \left(a^2 + b^2\right) * c = \left(a^2 + b^2\right)^2 + c^2 \tag{1.3.8.1}$$

$$\text{RHS:} \quad a * (b * c) = a * \left(b^2 + c^2\right) = a^2 + \left(b^2 + c^2\right)^2 \tag{1.3.8.2}$$

Choosing counterexample $a=1, b=2, c=3 \in \mathbb{R}$:

$$(1*2)*3 = \left(1^2+2^2\right)*3 = 5*3 = 5^2+3^2 = 25+9 = 34 \tag{1.3.8.3}$$

$$1*(2*3) = 1*\left(2^2+3^2\right) = 1*13 = 1^2+13^2 = 1+169 = 170 \tag{1.3.8.4}$$

Since $34 \neq 170$, the operation $*$ is **not associative**.

1.3.9 If $\|\mathbf{a}\| = 2$, $\|\mathbf{b}\| = 7$, and $\mathbf{a}\times\mathbf{b} = \begin{pmatrix}3\\2\\6\end{pmatrix}$, find the angle between $\mathbf{a}$ and $\mathbf{b}$. (12 2019)

Solution: Let θ be the angle between $\mathbf{a}$ and $\mathbf{b}$. Evaluating the magnitude of $\mathbf{a}\times\mathbf{b}$:

$$\|\mathbf{a}\times\mathbf{b}\| = \sqrt{3^2+2^2+6^2} = \sqrt{9+4+36} = \sqrt{49} = 7 \tag{1.3.9.1}$$

From (1.4.9.1) and (1.4.9.2):

$$\sin\theta = \frac{\|\mathbf{a}\times\mathbf{b}\|}{\|\mathbf{a}\|\,\|\mathbf{b}\|} = \frac{7}{2\times 7} = \frac{1}{2} \tag{1.3.9.2}$$

Since $\theta \in [0,\pi]$:

$$\theta = \frac{\pi}{6} \quad \text{or} \quad \theta = \frac{5\pi}{6} \tag{1.3.9.3}$$

1.3.10 Find the volume of a cuboid whose edges are given by $\mathbf{a} = \begin{pmatrix}-3\\7\\5\end{pmatrix}$, $\mathbf{b} = \begin{pmatrix}-5\\7\\-3\end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix}7\\-5\\-3\end{pmatrix}$. (12 2019)

Solution: From (1.4.10.1), the volume V of a parallelepiped (cuboid) is the absolute value of the scalar triple product:

$$V = \left|\begin{matrix}\mathbf{a} & \mathbf{b} & \mathbf{c}\end{matrix}\right| = \left\|\begin{vmatrix}-3 & -5 & 7\\ 7 & 7 & -5\\ 5 & -3 & -3\end{vmatrix}\right\| \tag{1.3.10.1}$$

Expanding the determinant:

$$\begin{vmatrix}\mathbf{a} & \mathbf{b} & \mathbf{c}\end{vmatrix} = -3\begin{vmatrix}7 & -5\\ -3 & -3\end{vmatrix} - (-5)\begin{vmatrix}7 & -5\\ 5 & -3\end{vmatrix} + 7\begin{vmatrix}7 & 7\\ 5 & -3\end{vmatrix} \tag{1.3.10.2}$$

$$= -3(-21-15) + 5(-21-(-25)) + 7(-21-35) \tag{1.3.10.3}$$

$$= -3(-36) + 5(4) + 7(-56) = 108+20-392 = -264 \tag{1.3.10.4}$$

Taking the absolute value:

$$V = |-264| = 264 \text{ cubic units} \tag{1.3.10.5}$$

The following code verifies the volume:

```
codes/12/q10.py
```

1.3.11 A coin is tossed 5 times. What is the probability of getting (i) 3 heads, (ii) at most

3 heads? (12 2019)

Solution: Let X denote the number of heads obtained in $n = 5$ independent coin tosses. This represents a sequence of Bernoulli trials with success probability $p = \frac{1}{2}$ and failure probability $q = 1 - p = \frac{1}{2}$. From the binomial distribution formula (1.4.11.1):

$$\Pr(X = k) = \binom{n}{k} p^k q^{n-k} = \binom{5}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{5-k} = \frac{\binom{5}{k}}{32} \tag{1.3.11.1}$$

(i) Probability of getting exactly 3 heads ($k = 3$):

$$\Pr(X = 3) = \frac{\binom{5}{3}}{32} = \frac{10}{32} = \frac{5}{16} \tag{1.3.11.2}$$

(ii) Probability of getting at most 3 heads ($\Pr(X \leq 3)$): Using the complement relation from (1.4.11.2):

$$\Pr(X \leq 3) = 1 - [\Pr(X = 4) + \Pr(X = 5)] \tag{1.3.11.3}$$

$$= 1 - \left(\frac{\binom{5}{4}}{32} + \frac{\binom{5}{5}}{32}\right) = 1 - \frac{5+1}{32} = 1 - \frac{6}{32} = \frac{26}{32} = \frac{13}{16} \tag{1.3.11.4}$$

1.3.12 Find the probability distribution of X, the number of heads in a simultaneous toss of two coins. (12 2019)

Solution: When two fair coins are tossed simultaneously, the sample space S is:

$$S = \{\text{HH, HT, TH, TT}\} \tag{1.3.12.1}$$

Let X be the random variable representing the number of heads. The possible values of X are $0, 1, 2$. From (1.4.12.1):

$$\Pr(X = 0) = \frac{n(\{\text{TT}\})}{n(S)} = \frac{1}{4} \tag{1.3.12.2}$$

$$\Pr(X = 1) = \frac{n(\{\text{HT, TH}\})}{n(S)} = \frac{2}{4} = \frac{1}{2} \tag{1.3.12.3}$$

$$\Pr(X = 2) = \frac{n(\{\text{HH}\})}{n(S)} = \frac{1}{4} \tag{1.3.12.4}$$

Verifying the distribution sum:

$$\sum_{x=0}^{2} \Pr(X = x) = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1 \tag{1.3.12.5}$$

1.3.13 Find $\displaystyle\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x}\, dx$. (12 2019)

Solution: Let $I = \int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x}\, dx$. Splitting the fraction into two terms:

$$\frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} = \frac{\sin^3 x}{\sin^2 x \cos^2 x} + \frac{\cos^3 x}{\sin^2 x \cos^2 x} \tag{1.3.13.1}$$

$$= \frac{\sin x}{\cos^2 x} + \frac{\cos x}{\sin^2 x} \tag{1.3.13.2}$$

$$= \left(\frac{\sin x}{\cos x}\right)\left(\frac{1}{\cos x}\right) + \left(\frac{\cos x}{\sin x}\right)\left(\frac{1}{\sin x}\right) \tag{1.3.13.3}$$

$$= \sec x \tan x + \csc x \cot x \tag{1.3.13.4}$$

Integrating term-by-term using (1.4.13.1) and (1.4.13.2):

$$I = \int (\sec x \tan x + \csc x \cot x)\, dx \tag{1.3.13.5}$$

$$= \int \sec x \tan x\, dx + \int \csc x \cot x\, dx \tag{1.3.13.6}$$

$$= \sec x - \csc x + C \tag{1.3.13.7}$$

1.3.14 Find $\int \frac{x-3}{(x-1)^3} e^x\, dx$. (12 2019)

Solution: Let $I = \int \frac{x-3}{(x-1)^3} e^x\, dx$. Expressing the numerator in terms of $(x-1)$:

$$\frac{x-3}{(x-1)^3} = \frac{(x-1)-2}{(x-1)^3} = \frac{1}{(x-1)^2} - \frac{2}{(x-1)^3} \tag{1.3.14.1}$$

Let $f(x) = \frac{1}{(x-1)^2} = (x-1)^{-2}$. Its derivative is:

$$f'(x) = -2(x-1)^{-3} = -\frac{2}{(x-1)^3} \tag{1.3.14.2}$$

Using the standard exponential integral formula (1.4.14.1):

$$\int e^x [f(x) + f'(x)]\, dx = e^x f(x) + C \tag{1.3.14.3}$$

Substituting $f(x)$:

$$I = \int e^x \left[\frac{1}{(x-1)^2} - \frac{2}{(x-1)^3}\right] dx = \frac{e^x}{(x-1)^2} + C \tag{1.3.14.4}$$

1.3.15 If $\Pr(\text{not } A) = 0.7$, $\Pr(B) = 0.7$, and $\Pr(B \mid A) = 0.5$, find $\Pr(A \mid B)$. (12 2019)

Solution: From the complement rule, $\Pr(A) = 1 - \Pr(\text{not } A) = 1 - 0.7 = 0.3$. From the definition of conditional probability (1.4.15.1):

$$\Pr(B \mid A) = \frac{\Pr(A \cap B)}{\Pr(A)} \implies 0.5 = \frac{\Pr(A \cap B)}{0.3} \implies \Pr(A \cap B) = 0.15 \tag{1.3.15.1}$$

Now, computing $\Pr(A \mid B)$ via (1.4.15.1):

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)} = \frac{0.15}{0.7} = \frac{15}{70} = \frac{3}{14} \approx 0.2143 \tag{1.3.15.2}$$

1.3.16 Find the general solution of the differential equation $\frac{dy}{dx} = e^{x+y}$. (12 2019)

Solution: Using index laws, the differential equation can be separated:

$$\frac{dy}{dx} = e^x e^y \implies e^{-y}\,dy = e^x\,dx \tag{1.3.16.1}$$

Integrating both sides using (1.4.16.1):

$$\int e^{-y}\,dy = \int e^x\,dx \implies -e^{-y} = e^x + C_1 \tag{1.3.16.2}$$

Rearranging terms:

$$e^x + e^{-y} = C \quad \text{where } C = -C_1 \tag{1.3.16.3}$$

1.3.17 Find the value of $\sin\left(\cos^{-1}\frac{4}{5} + \tan^{-1}\frac{2}{3}\right)$. (12 2019)

Solution: From (1.4.17.1), converting $\cos^{-1}\left(\frac{4}{5}\right)$ to $\tan^{-1}$:

$$\cos^{-1}\frac{4}{5} = \tan^{-1}\left(\frac{\sqrt{1-(4/5)^2}}{4/5}\right) = \tan^{-1}\left(\frac{3/5}{4/5}\right) = \tan^{-1}\frac{3}{4} \tag{1.3.17.1}$$

Using the addition formula (1.4.17.2) for $\tan^{-1}\frac{3}{4} + \tan^{-1}\frac{2}{3}$ (since $uv = \frac{3}{4}\cdot\frac{2}{3} = \frac{1}{2} < 1$):

$$\tan^{-1}\frac{3}{4} + \tan^{-1}\frac{2}{3} = \tan^{-1}\left(\frac{\frac{3}{4}+\frac{2}{3}}{1-\frac{3}{4}\cdot\frac{2}{3}}\right) \tag{1.3.17.2}$$

$$= \tan^{-1}\left(\frac{\frac{17}{12}}{1-\frac{6}{12}}\right) = \tan^{-1}\left(\frac{17/12}{6/12}\right) = \tan^{-1}\frac{17}{6} \tag{1.3.17.3}$$

Let $\theta = \tan^{-1}\frac{17}{6}$. From (1.4.17.3):

$$\sin\left(\tan^{-1}\frac{17}{6}\right) = \frac{17}{\sqrt{17^2+6^2}} = \frac{17}{\sqrt{289+36}} = \frac{17}{\sqrt{325}} = \frac{17}{5\sqrt{13}} = \frac{17\sqrt{13}}{65} \tag{1.3.17.4}$$

1.3.18 Using properties of determinants, show that

$$\begin{vmatrix} 3a & -a+b & -a+c \\ -b+a & 3b & -b+c \\ -c+a & -c+b & 3c \end{vmatrix} = 3(a+b+c)(ab+bc+ca) \tag{1.3.18.1}$$

(12 2019)

Solution: Let Δ denote the given determinant. Applying the column operation $C_1 \to$

$C_1 + C_2 + C_3$ in accordance with (1.4.18.1):

$$\Delta = \begin{vmatrix} a+b+c & -a+b & -a+c \\ a+b+c & 3b & -b+c \\ a+b+c & -c+b & 3c \end{vmatrix} \tag{1.3.18.2}$$

Factoring out $(a+b+c)$ from column C_1:

$$\Delta = (a+b+c) \begin{vmatrix} 1 & -a+b & -a+c \\ 1 & 3b & -b+c \\ 1 & -c+b & 3c \end{vmatrix} \tag{1.3.18.3}$$

Applying row operations $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$:

$$\Delta = (a+b+c) \begin{vmatrix} 1 & -a+b & -a+c \\ 0 & a+2b & a-b \\ 0 & a-c & a+2c \end{vmatrix} \tag{1.3.18.4}$$

Expanding along column C_1:

$$\Delta = (a+b+c) \begin{vmatrix} a+2b & a-b \\ a-c & a+2c \end{vmatrix} \tag{1.3.18.5}$$

$$= (a+b+c)\left[(a+2b)(a+2c) - (a-b)(a-c)\right] \tag{1.3.18.6}$$

$$= (a+b+c)\left[\left(a^2+2ac+2ab+4bc\right) - \left(a^2-ac-ab+bc\right)\right] \tag{1.3.18.7}$$

$$= (a+b+c)(3ab+3bc+3ca) \tag{1.3.18.8}$$

$$= 3(a+b+c)(ab+bc+ca) \tag{1.3.18.9}$$

1.3.19 Check whether the relation R defined on the set $A = \{1,2,3,4,5,6\}$ by $R = \{(a,b) : b = a+1\}$ is reflexive, symmetric or transitive. (12 2019)

Solution:

Listing all ordered pairs satisfying $b = a+1$ for $a, b \in A$:

$$R = \{(1,2), (2,3), (3,4), (4,5), (5,6)\} \tag{1.3.19.1}$$

- **Reflexivity:** A relation is reflexive if $(a,a) \in R$ for all $a \in A$. Since $(1,1) \notin R$, the relation is **not reflexive**.
- **Symmetry:** A relation is symmetric if $(a,b) \in R \implies (b,a) \in R$. Since $(1,2) \in R$ but $(2,1) \notin R$, the relation is **not symmetric**.
- **Transitivity:** A relation is transitive if $(a,b) \in R$ and $(b,c) \in R \implies (a,c) \in R$. Since $(1,2) \in R$ and $(2,3) \in R$ but $(1,3) \notin R$, the relation is **not transitive**.

Thus, R is neither reflexive, nor symmetric, nor transitive.

1.3.20 Let $f : \mathbb{N} \to Y$ be a function defined as $f(x) = 4x+3$, where $Y = \{y \in \mathbb{N} : y = 4x+3 \text{ for some } x \in \mathbb{N}\}$. Show that f is invertible. Find its inverse. (12 2019)

Solution: From (1.4.20.1), to prove f is injective (one-one), let $f(x_1) = f(x_2)$ for $x_1, x_2 \in \mathbb{N}$:

$$4x_1 + 3 = 4x_2 + 3 \implies 4x_1 = 4x_2 \implies x_1 = x_2 \tag{1.3.20.1}$$

Thus, f is injective. From (1.4.20.2), for any $y \in Y$, by definition $y = 4x+3$ for

some $x \in \mathbb{N}$. Solving for x:

$$x = \frac{y-3}{4} \in \mathbb{N} \tag{1.3.20.2}$$

Since every $y \in Y$ has a pre-image in $\mathbb{N}$, f is surjective (onto). Since f is both injective and surjective, it is bijective and hence invertible. Defining the inverse function $f^{-1} : Y \to \mathbb{N}$:

$$f^{-1}(y) = \frac{y-3}{4} \tag{1.3.20.3}$$

1.3.21 If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, where $x \neq y$, prove that $\dfrac{dy}{dx} = -\dfrac{1}{(x+1)^2}$. (12 2019)

Solution: Rearranging the given equation:

$$x\sqrt{1+y} = -y\sqrt{1+x} \tag{1.3.21.1}$$

Squaring both sides:

$$x^2(1+y) = y^2(1+x) \tag{1.3.21.2}$$
$$x^2 + x^2y = y^2 + xy^2 \tag{1.3.21.3}$$
$$x^2 - y^2 + x^2y - xy^2 = 0 \tag{1.3.21.4}$$
$$(x-y)(x+y) + xy(x-y) = 0 \tag{1.3.21.5}$$
$$(x-y)[x+y+xy] = 0 \tag{1.3.21.6}$$

Since $x \neq y$, we have $x - y \neq 0$, which implies:

$$x + y + xy = 0 \implies y(1+x) = -x \implies y = -\frac{x}{1+x} \tag{1.3.21.7}$$

Rewriting y to facilitate differentiation:

$$y = -\frac{(x+1)-1}{x+1} = -\left(1 - \frac{1}{x+1}\right) = -1 + (x+1)^{-1} \tag{1.3.21.8}$$

Differentiating with respect to x:

$$\frac{dy}{dx} = \frac{d}{dx}\left[-1 + (x+1)^{-1}\right] = -(x+1)^{-2} \cdot 1 = -\frac{1}{(x+1)^2} \tag{1.3.21.9}$$

1.3.22 If $(\cos x)^y = (\sin y)^x$, find $\dfrac{dy}{dx}$. (12 2019)

Solution: Taking the natural logarithm on both sides using (1.4.21.1):

$$\ln\left((\cos x)^y\right) = \ln\left((\sin y)^x\right) \tag{1.3.22.1}$$
$$y \ln(\cos x) = x \ln(\sin y) \tag{1.3.22.2}$$

Differentiating both sides implicitly with respect to x using (1.4.21.2):

$$\frac{dy}{dx}\ln(\cos x)+y\cdot\frac{-\sin x}{\cos x}=1\cdot\ln(\sin y)+x\cdot\frac{\cos y}{\sin y}\frac{dy}{dx} \tag{1.3.22.3}$$

$$\frac{dy}{dx}\ln(\cos x)-y\tan x=\ln(\sin y)+x\cot y\frac{dy}{dx} \tag{1.3.22.4}$$

Grouping the derivative terms:

$$\frac{dy}{dx}\left[\ln(\cos x)-x\cot y\right]=\ln(\sin y)+y\tan x \tag{1.3.22.5}$$

Solving for $\dfrac{dy}{dx}$:

$$\frac{dy}{dx}=\frac{\ln(\sin y)+y\tan x}{\ln(\cos x)-x\cot y} \tag{1.3.22.6}$$

1.3.23 Prove that $\displaystyle\int_0^a f(x)\,dx=\int_0^a f(a-x)\,dx$, and hence evaluate $\displaystyle\int_0^{\pi/2}\frac{x}{\sin x+\cos x}\,dx$. (12 2019)

Solution: To prove King's property (1.4.22.1), let $t=a-x\implies dx=-dt$. When $x=0$, $t=a$; when $x=a$, $t=0$:

$$\int_0^a f(x)\,dx=\int_a^0 f(a-t)\,(-dt)=\int_0^a f(a-t)\,dt=\int_0^a f(a-x)\,dx \tag{1.3.23.1}$$

Now, evaluating the given definite integral:

$$I=\int_0^{\pi/2}\frac{x}{\sin x+\cos x}\,dx \tag{1.3.23.2}$$

Applying property (1.4.22.1):

$$I=\int_0^{\pi/2}\frac{\frac{\pi}{2}-x}{\sin\left(\frac{\pi}{2}-x\right)+\cos\left(\frac{\pi}{2}-x\right)}\,dx \tag{1.3.23.3}$$

$$=\int_0^{\pi/2}\frac{\frac{\pi}{2}-x}{\cos x+\sin x}\,dx \tag{1.3.23.4}$$

Adding (1.3.23.2) and (1.3.23.4):

$$2I=\int_0^{\pi/2}\frac{x+\frac{\pi}{2}-x}{\sin x+\cos x}\,dx=\frac{\pi}{2}\int_0^{\pi/2}\frac{1}{\sin x+\cos x}\,dx \tag{1.3.23.5}$$

$$I=\frac{\pi}{4}\int_0^{\pi/2}\frac{1}{\sqrt{2}\cos\left(x-\frac{\pi}{4}\right)}\,dx=\frac{\pi}{4\sqrt{2}}\int_0^{\pi/2}\sec\left(x-\frac{\pi}{4}\right)\,dx \tag{1.3.23.6}$$

Substituting $u = x - \frac{\pi}{4}$:

$$I = \frac{\pi}{4\sqrt{2}} \int_{-\pi/4}^{\pi/4} \sec u \, du = \frac{\pi}{4\sqrt{2}} \cdot 2 \ln \left| \sec \frac{\pi}{4} + \tan \frac{\pi}{4} \right| \tag{1.3.23.7}$$

$$= \frac{\pi}{2\sqrt{2}} \ln \left(\sqrt{2} + 1 \right) \tag{1.3.23.8}$$

1.3.24 Find $\displaystyle\int \frac{x^2 + x + 1}{(x+2)(x^2+1)} \, dx$. (12 2019)

Solution: Using partial fraction decomposition from (1.4.23.1):

$$\frac{x^2 + x + 1}{(x+2)(x^2+1)} = \frac{A}{x+2} + \frac{Bx + C}{x^2+1} \tag{1.3.24.1}$$

Multiplying by $(x+2)(x^2+1)$:

$$x^2 + x + 1 = A(x^2+1) + (Bx + C)(x+2) \tag{1.3.24.2}$$

$$= (A+B)x^2 + (2B+C)x + (A+2C) \tag{1.3.24.3}$$

Comparing coefficients of like powers of x:

$$A + B = 1 \tag{1.3.24.4}$$

$$2B + C = 1 \tag{1.3.24.5}$$

$$A + 2C = 1 \tag{1.3.24.6}$$

Solving this linear system yields $A = \frac{3}{5}$, $B = \frac{2}{5}$, and $C = \frac{1}{5}$. Thus:

$$\frac{x^2 + x + 1}{(x+2)(x^2+1)} = \frac{3}{5}\left(\frac{1}{x+2}\right) + \frac{1}{5}\left(\frac{2x}{x^2+1}\right) + \frac{1}{5}\left(\frac{1}{x^2+1}\right) \tag{1.3.24.7}$$

Integrating term-by-term using (1.4.23.2)–(1.4.23.4):

$$I = \frac{3}{5}\int \frac{1}{x+2} \, dx + \frac{1}{5}\int \frac{2x}{x^2+1} \, dx + \frac{1}{5}\int \frac{1}{x^2+1} \, dx \tag{1.3.24.8}$$

$$= \frac{3}{5}\ln|x+2| + \frac{1}{5}\ln\left(x^2+1\right) + \frac{1}{5}\tan^{-1} x + C \tag{1.3.24.9}$$

1.3.25 Solve the differential equation $x\dfrac{dy}{dx} = y - x\tan\left(\frac{y}{x}\right)$. (12 2019)

Solution: Rewriting in the homogeneous form (1.4.24.1):

$$\frac{dy}{dx} = \frac{y}{x} - \tan\left(\frac{y}{x}\right) \tag{1.3.25.1}$$

Substituting $y = vx$ and $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$ in accordance with (1.4.24.2):

$$v + x\frac{dv}{dx} = v - \tan v \implies x\frac{dv}{dx} = -\tan v \tag{1.3.25.2}$$

Separating variables:

$$\frac{dv}{\tan v} = -\frac{dx}{x} \implies \cot v\, dv = -\frac{dx}{x} \tag{1.3.25.3}$$

Integrating both sides:

$$\ln|\sin v| = -\ln|x| + \ln C_1 \implies \ln|x \sin v| = \ln C_1 \implies x \sin v = C \tag{1.3.25.4}$$

Substituting back $v = \frac{y}{x}$:

$$x \sin\left(\frac{y}{x}\right) = C \tag{1.3.25.5}$$

1.3.26 Solve the differential equation $\dfrac{dy}{dx} = -\dfrac{x + y\cos x}{1 + \sin x}$. (12 2019)

Solution: Rearranging into the standard linear differential form (1.4.25.1):

$$\frac{dy}{dx} + \left(\frac{\cos x}{1+\sin x}\right) y = -\frac{x}{1+\sin x} \tag{1.3.26.1}$$

Here $P(x) = \frac{\cos x}{1+\sin x}$ and $Q(x) = -\frac{x}{1+\sin x}$. Finding the integrating factor via (1.4.25.3):

$$\text{I.F.} = e^{\int \frac{\cos x}{1+\sin x}\, dx} = e^{\ln(1+\sin x)} = 1 + \sin x \tag{1.3.26.2}$$

Multiplying through and integrating:

$$y\,(1+\sin x) = \int \left(-\frac{x}{1+\sin x}\right)(1+\sin x)\, dx + C \tag{1.3.26.3}$$

$$= \int (-x)\, dx + C = -\frac{x^2}{2} + C \tag{1.3.26.4}$$

Multiplying by 2:

$$2y\,(1+\sin x) + x^2 = C' \tag{1.3.26.5}$$

1.3.27 The scalar product of the vector $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ with a unit vector along the sum of the vectors $\mathbf{b} = 2\mathbf{i} + 4\mathbf{j} - 5\mathbf{k}$ and $\mathbf{c} = \lambda\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ is equal to 1. Find the value of λ and hence find the unit vector along $\mathbf{b} + \mathbf{c}$. (12 2019)

Solution: Summing vectors $\mathbf{b}$ and $\mathbf{c}$:

$$\mathbf{b} + \mathbf{c} = \begin{pmatrix} 2+\lambda \\ 6 \\ -2 \end{pmatrix} \tag{1.3.27.1}$$

The unit vector $\mathbf{u}$ along $\mathbf{b} + \mathbf{c}$ is given by (1.4.26.1):

$$\mathbf{u} = \frac{\mathbf{b}+\mathbf{c}}{\|\mathbf{b}+\mathbf{c}\|} = \frac{\begin{pmatrix} \lambda+2 \\ 6 \\ -2 \end{pmatrix}}{\sqrt{(\lambda+2)^2 + 36 + 4}} = \frac{\begin{pmatrix} \lambda+2 \\ 6 \\ -2 \end{pmatrix}}{\sqrt{(\lambda+2)^2 + 40}} \tag{1.3.27.2}$$

Given that $\mathbf{a}\cdot\mathbf{u}=1$:

$$\begin{pmatrix}1 & 1 & 1\end{pmatrix}\cdot\frac{\begin{pmatrix}\lambda+2\\6\\-2\end{pmatrix}}{\sqrt{(\lambda+2)^2+40}}=1\implies\frac{(\lambda+2)+6-2}{\sqrt{(\lambda+2)^2+40}}=1\implies\frac{\lambda+6}{\sqrt{(\lambda+2)^2+40}}=1 \tag{1.3.27.3}$$

Squaring both sides ($\lambda+6>0$):

$$(\lambda+6)^2=(\lambda+2)^2+40\implies\lambda^2+12\lambda+36=\lambda^2+4\lambda+4+40\implies 8\lambda=8\implies\lambda \tag{1.3.27.4}$$

Substituting $\lambda=1$:

$$\mathbf{b}+\mathbf{c}=\begin{pmatrix}3\\6\\-2\end{pmatrix},\quad\|\mathbf{b}+\mathbf{c}\|=\sqrt{9+36+4}=\sqrt{49}=7 \tag{1.3.27.5}$$

Thus, the unit vector along $\mathbf{b}+\mathbf{c}$ is:

$$\mathbf{u}=\frac{1}{7}\begin{pmatrix}3\\6\\-2\end{pmatrix}=\frac{3}{7}\mathbf{i}+\frac{6}{7}\mathbf{j}-\frac{2}{7}\mathbf{k} \tag{1.3.27.6}$$

1.3.28 If $(a+bx)\,e^{y/x}=x$, then prove that $x^3\dfrac{d^2y}{dx^2}=\left(x\dfrac{dy}{dx}-y\right)^2$. (12 2019)

Solution: Taking the natural logarithm on both sides:

$$\ln(a+bx)+\frac{y}{x}=\ln x\implies\frac{y}{x}=\ln x-\ln(a+bx) \tag{1.3.28.1}$$

Differentiating with respect to x:

$$\frac{x\frac{dy}{dx}-y}{x^2}=\frac{1}{x}-\frac{b}{a+bx}=\frac{a+bx-bx}{x(a+bx)}=\frac{a}{x(a+bx)} \tag{1.3.28.2}$$

Multiplying by x^2:

$$x\frac{dy}{dx}-y=\frac{ax}{a+bx} \tag{1.3.28.3}$$

Differentiating (1.3.28.3) with respect to x:

$$\frac{dy}{dx}+x\frac{d^2y}{dx^2}-\frac{dy}{dx}=\frac{(a+bx)(a)-(ax)(b)}{(a+bx)^2} \tag{1.3.28.4}$$

$$x\frac{d^2y}{dx^2}=\frac{a^2}{(a+bx)^2} \tag{1.3.28.5}$$

Multiplying both sides by x^2:

$$x^3\frac{d^2y}{dx^2} = \frac{a^2x^2}{(a+bx)^2} = \left(\frac{ax}{a+bx}\right)^2 \tag{1.3.28.6}$$

Substituting (1.3.28.3):

$$x^3\frac{d^2y}{dx^2} = \left(x\frac{dy}{dx} - y\right)^2 \tag{1.3.28.7}$$

The following code verifies the differential identity:

codes/12/q28.py

1.3.29 The volume of a cube is increasing at the rate of 8 cm^3/s. How fast is the surface area increasing when the length of its edge is 12 cm? (12 2019)

Solution: Let x be the edge length of the cube. From (1.4.27.2):

$$V = x^3 \implies \frac{dV}{dt} = 3x^2\frac{dx}{dt} \tag{1.3.29.1}$$

Given $\frac{dV}{dt} = 8$ cm^3/s:

$$8 = 3x^2\frac{dx}{dt} \implies \frac{dx}{dt} = \frac{8}{3x^2} \tag{1.3.29.2}$$

Surface area $S = 6x^2$. From (1.4.27.2), the rate of change of surface area is:

$$\frac{dS}{dt} = 12x\frac{dx}{dt} = 12x\left(\frac{8}{3x^2}\right) = \frac{32}{x} \tag{1.3.29.3}$$

Evaluating at $x = 12$ cm:

$$\left.\frac{dS}{dt}\right|_{x=12} = \frac{32}{12} = \frac{8}{3}\ \text{cm}^2/\text{s} \tag{1.3.29.4}$$

1.3.30 Find the Cartesian and vector equations of the plane passing through the points $A(2,5,-3)$, $B(-2,-3,5)$, and $C(5,3,-3)$. (12 2019)

Solution: Let the position vectors of the points be

$$\mathbf{a} = \begin{pmatrix}2\\5\\-3\end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix}-2\\-3\\5\end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix}5\\3\\-3\end{pmatrix} \tag{1.3.30.1}$$

Evaluating displacement vectors lying in the plane:

$$\mathbf{AB} = \mathbf{b} - \mathbf{a} = \begin{pmatrix}-4\\-8\\8\end{pmatrix} \tag{1.3.30.2}$$

$$\mathbf{AC} = \mathbf{c} - \mathbf{a} = \begin{pmatrix}3\\-2\\0\end{pmatrix} \tag{1.3.30.3}$$

From (1.4.28.2), the normal vector $\mathbf{n}$ is obtained by cross product:

$$\mathbf{n} = \mathbf{AB} \times \mathbf{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -4 & -8 & 8 \\ 3 & -2 & 0 \end{vmatrix} = \mathbf{i}\left(0-(-16)\right) - \mathbf{j}\left(0-24\right) + \mathbf{k}\left(8-(-24)\right) = 16\mathbf{i} + 2 \tag{1.3.30.4}$$

Scaling by $\frac{1}{8}$ yields the simplified normal vector:

$$\mathbf{n}_0 = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \tag{1.3.30.5}$$

From (1.4.28.2), evaluating $\mathbf{a} \cdot \mathbf{n}_0$:

$$\begin{pmatrix} 2 & 5 & -3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = 4 + 15 - 12 = 7 \tag{1.3.30.6}$$

Thus, the vector equation of the plane is:

$$\mathbf{r} \cdot \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = 7 \tag{1.3.30.7}$$

Substituting $\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, the Cartesian equation is:

$$2x + 3y + 4z = 7 \tag{1.3.30.8}$$

The following code verifies the plane equations:

```
codes/12/q30.py
```

1.3.31 Find the point on the curve $y^2 = 4x$ which is nearest to the point $(2, -8)$. (12 2019)

Solution: Let $P\left(x, y\right)$ be a point on the parabola $y^2 = 4x$, so its coordinates can be expressed parametrically as $\begin{pmatrix} y^2/4 \\ y \end{pmatrix}$. Let the fixed point be $\mathbf{A} = \begin{pmatrix} 2 \\ -8 \end{pmatrix}$. The square of the Euclidean distance Z between P and A from (1.4.29.1) is:

$$Z = \left(\frac{y^2}{4} - 2\right)^2 + \left(y - (-8)\right)^2 = \left(\frac{y^2}{4} - 2\right)^2 + \left(y + 8\right)^2 \tag{1.3.31.1}$$

Differentiating Z with respect to y:

$$\frac{dZ}{dy} = 2\left(\frac{y^2}{4} - 2\right)\left(\frac{2y}{4}\right) + 2\left(y + 8\right) \tag{1.3.31.2}$$

$$= \frac{y^3}{4} - 2y + 2y + 16 = \frac{y^3}{4} + 16 \tag{1.3.31.3}$$

Setting $\frac{dZ}{dy} = 0$ for stationary points:

$$\frac{y^3}{4} + 16 = 0 \implies y^3 = -64 \implies y = -4 \tag{1.3.31.4}$$

Applying the second derivative test (1.4.29.2):

$$\frac{d^2Z}{dy^2} = \frac{3y^2}{4} \tag{1.3.31.5}$$

Evaluating at $y = -4$:

$$\left.\frac{d^2Z}{dy^2}\right|_{y=-4} = \frac{3\left(-4\right)^2}{4} = \frac{3\left(16\right)}{4} = 12 > 0 \tag{1.3.31.6}$$

Since $\frac{d^2Z}{dy^2} > 0$, Z attains its minimum at $y = -4$. Substituting $y = -4$ into $x = \frac{y^2}{4}$:

$$x = \frac{\left(-4\right)^2}{4} = \frac{16}{4} = 4 \tag{1.3.31.7}$$

Therefore, the point on the curve nearest to $(2, -8)$ is $(4, -4)$. The following code verifies the optimization:

codes/12/q31.py

1.3.32 Find $\int_1^3 \left(x^2 + 2 + e^{2x}\right)\,dx$ as the limit of sums. (12 2019)

Solution: From the Riemann sum definition (1.4.30.1):

$$\int_a^b f(x)\,dx = \lim_{h\to 0} h\sum_{r=1}^{n} f\left(a + rh\right) \tag{1.3.32.1}$$

where $a = 1, b = 3 \implies nh = b - a = 2$, and $f(x) = x^2 + 2 + e^{2x}$. Evaluating $f\left(1 + rh\right)$:

$$f\left(1 + rh\right) = \left(1 + rh\right)^2 + 2 + e^{2(1+rh)} = 3 + 2rh + r^2h^2 + e^2\left(e^{2h}\right)^r \tag{1.3.32.2}$$

Using standard summation formulas (1.4.30.2), (1.4.30.3), and (1.4.30.4):

$$hS_n = 3\left(nh\right) + 2h^2\left[\frac{n\left(n+1\right)}{2}\right] + h^3\left[\frac{n\left(n+1\right)\left(2n+1\right)}{6}\right] + e^2e^{2h}\left(\frac{e^{2nh}-1}{e^{2h}-1}\right)h \tag{1.3.32.3}$$

$$= 3\left(nh\right) + \left(nh\right)\left(nh+h\right) + \frac{\left(nh\right)\left(nh+h\right)\left(2nh+h\right)}{6} + e^2\left(e^{2nh}-1\right)e^{2h}\left(\frac{2h}{e^{2h}-1}\right) \tag{1.3.32.4}$$

Taking the limit as $h \to 0$ using (1.4.30.5):

$$\int_1^3 \left(x^2 + 2 + e^{2x}\right) dx = 3\,(2) + 2\,(2) + \frac{2\,(2)\,(4)}{6} + \left(e^6 - e^2\right)(1)\left(\frac{1}{2} \cdot 1\right) \tag{1.3.32.5}$$

$$= 6 + 4 + \frac{8}{3} + \frac{e^6 - e^2}{2} = \frac{38}{3} + \frac{e^6 - e^2}{2} \tag{1.3.32.6}$$

1.3.33 Using integration, find the area of the triangular region whose sides have the equations $y = 2x + 1$, $y = 3x + 1$, and $x = 4$. (12 2019)

Solution: Solving the lines simultaneously:

- Intersection of $y = 2x + 1$ and $y = 3x + 1 \implies x = 0, y = 1$ (Vertex $A(0, 1)$).
- Intersection of $y = 2x + 1$ and $x = 4 \implies x = 4, y = 9$ (Vertex $B(4, 9)$).
- Intersection of $y = 3x + 1$ and $x = 4 \implies x = 4, y = 13$ (Vertex $C(4, 13)$).

From (1.4.31.1), integrating between $x = 0$ and $x = 4$:

$$\text{Area} = \int_0^4 \left[(3x + 1) - (2x + 1)\right] dx = \int_0^4 x\, dx \tag{1.3.33.1}$$

$$= \frac{x^2}{2}\Big|_0^4 = \frac{4^2}{2} = 8 \text{ square units} \tag{1.3.33.2}$$

The following code verifies the area:

```
codes/12/q33.py
```

1.3.34 If $\mathbf{A} = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 1 & 2 \\ 5 & 1 & 1 \end{bmatrix}$, find $\mathbf{A}^{-1}$. Hence solve the system of equations:

$$x + 3y + 4z = 8 \tag{1.3.34.1}$$

$$2x + y + 2z = 5 \tag{1.3.34.2}$$

$$5x + y + z = 7 \tag{1.3.34.3}$$

(12 2019)

Solution: Evaluating the determinant of $\mathbf{A}$:

$$\left|\mathbf{A}\right| = 1\,(1 - 2) - 3\,(2 - 10) + 4\,(2 - 5) = 1\,(-1) - 3\,(-8) + 4\,(-3) = -1 + 24 - 12 = 1 \tag{1.3.34.4}$$

Using (1.4.32.1), computing the cofactor matrix and its transpose (adjugate):

$$\text{adj}(\mathbf{A}) = \begin{bmatrix} -1 & 1 & 2 \\ 8 & -19 & 6 \\ -3 & 14 & -5 \end{bmatrix} \tag{1.3.34.5}$$

Thus, $\mathbf{A}^{-1}$ is given by:

$$\mathbf{A}^{-1} = \frac{1}{11}\begin{bmatrix} -1 & 1 & 2 \\ 8 & -19 & 6 \\ -3 & 14 & -5 \end{bmatrix} \tag{1.3.34.6}$$

Writing the system of equations in matrix form $\mathbf{AX} = \mathbf{B}$ where $\mathbf{X} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 8 \\ 5 \\ 7 \end{pmatrix}$:

$$\mathbf{X} = \mathbf{A}^{-1}\mathbf{B} = \frac{1}{11}\begin{bmatrix} -1 & 1 & 2 \\ 8 & -19 & 6 \\ -3 & 14 & -5 \end{bmatrix}\begin{pmatrix} 8 \\ 5 \\ 7 \end{pmatrix} = \frac{1}{11}\begin{pmatrix} -8+5+14 \\ 64-95+42 \\ -24+70-35 \end{pmatrix} = \frac{1}{11}\begin{pmatrix} 11 \\ 11 \\ 11 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \tag{1.3.34.7}$$

Therefore, $x = 1$, $y = 1$, and $z = 1$.

1.3.35 Find the inverse of the matrix $\mathbf{A} = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$ using elementary transformations. (12 2019)

Solution: From (1.4.33.1), setting up the augmented matrix $[\mathbf{A} \mid \mathbf{I}]$ and applying Gauss-Jordan row elimination:

$$[\mathbf{A} \mid \mathbf{I}] = \left[\begin{array}{ccc|ccc} 2 & 0 & -1 & 1 & 0 & 0 \\ 5 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 3 & 0 & 0 & 1 \end{array}\right] \tag{1.3.35.1}$$

Performing elementary row transformations $R_2 \to R_2 - 2R_1$, row swaps, and elimination steps reduces the left block to the identity matrix $\mathbf{I}$, yielding:

$$\mathbf{A}^{-1} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix} \tag{1.3.35.2}$$

1.3.36 A company produces two types of goods, A and B, that require gold and silver. Each unit of type A requires 3 g of silver and 1 g of gold while that of type B requires 1 g of silver and 2 g of gold. The company can use at most 9 g of silver and 8 g of gold. If each unit of type A brings a profit of Rs. 40 and that of type B Rs. 50, find the number of units of each type that the company should produce to maximize profit. Formulate the above LPP and solve it graphically and find the maximum profit. (12 2019)

Solution: Let x and y be the units of type A and type B goods produced. From

(1.4.34.1) and (1.4.34.2), the LPP formulation is:

$$\text{Maximize } Z = 40x + 50y \tag{1.3.36.1}$$
$$\text{subject to } 3x + y \leq 9 \tag{1.3.36.2}$$
$$x + 2y \leq 8 \tag{1.3.36.3}$$
$$x \geq 0,\ y \geq 0 \tag{1.3.36.4}$$

The feasible region is the polygon $OABC$ bounded by the extreme corner points:

$$O(0,0), \quad A(3,0), \quad B(2,3), \quad C(0,4) \tag{1.3.36.5}$$

Evaluating the objective function Z at each corner point:

- At $O(0,0) : Z = 0$
- At $A(3,0) : Z = 40\,(3) + 50\,(0) = 120$
- At $B(2,3) : Z = 40\,(2) + 50\,(3) = 80 + 150 = 230$
- At $C(0,4) : Z = 40\,(0) + 50\,(4) = 200$

By the Corner-Point Theorem, the maximum profit is Rs. 230, achieved by producing 2 units of type A and 3 units of type B. The following code verifies the graphical optimization:

codes/12/q36.py

1.3.37 There are three coins. One is a two-headed coin, another is a biased coin that comes up heads 75% of the time and the third is an unbiased coin. One of the three coins is chosen at random and tossed. If it shows heads, what is the probability that it is the two-headed coin? (12 2019)

Solution: Let E_1, E_2, E_3 be the events of choosing the two-headed coin, the biased coin, and the unbiased coin respectively.

$$\Pr(E_1) = \Pr(E_2) = \Pr(E_3) = \frac{1}{3} \tag{1.3.37.1}$$

Let A be the event that the tossed coin shows heads. The conditional probabilities are:

$$\Pr(A \mid E_1) = 1, \quad \Pr(A \mid E_2) = \frac{75}{100} = \frac{3}{4}, \quad \Pr(A \mid E_3) = \frac{1}{2} \tag{1.3.37.2}$$

Using Bayes' theorem (1.4.35.2):

$$\Pr(E_1 \mid A) = \frac{\Pr(E_1)\Pr(A \mid E_1)}{\Pr(E_1)\Pr(A \mid E_1) + \Pr(E_2)\Pr(A \mid E_2) + \Pr(E_3)\Pr(A \mid E_3)} \tag{1.3.37.3}$$

$$= \frac{\frac{1}{3}(1)}{\frac{1}{3}(1) + \frac{1}{3}\left(\frac{3}{4}\right) + \frac{1}{3}\left(\frac{1}{2}\right)} = \frac{1}{1 + \frac{3}{4} + \frac{1}{2}} = \frac{1}{\frac{9}{4}} = \frac{4}{9} \tag{1.3.37.4}$$

1.3.38 Find the equation of the line passing through $(2, -1, 2)$ and $(5, 3, 4)$ and of the plane passing through $(2, 0, 3)$, $(1, 1, 5)$ and $(3, 2, 4)$. Also, find their point of intersection. (12 2019)

Solution: The direction ratios of the line passing through $(2,-1,2)$ and $(5,3,4)$ are $3,4,2$. Its parametric Cartesian equation from (1.4.5.2) is:

$$\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{2}=\lambda \implies x=3\lambda+2,\ y=4\lambda-1,\ z=2\lambda+2 \tag{1.3.38.1}$$

The plane passing through $(2,0,3)$, $(1,1,5)$, and $(3,2,4)$ has normal $\mathbf{n}=\begin{pmatrix}-1\\1\\2\end{pmatrix}\times\begin{pmatrix}1\\2\\1\end{pmatrix}=\begin{pmatrix}-3\\3\\-3\end{pmatrix}$, yielding normal vector $\mathbf{n}_0=\begin{pmatrix}1\\-1\\1\end{pmatrix}$. From (1.4.28.1), the Cartesian equation of the plane is:

$$x-y+z=5 \tag{1.3.38.2}$$

From (1.4.37.1), substituting the line parameterization into the plane equation:

$$(3\lambda+2)-(4\lambda-1)+(2\lambda+2)=5 \implies \lambda+5=5 \implies \lambda=0 \tag{1.3.38.3}$$

Substituting $\lambda=0$ into the line equations yields the point of intersection:

$$\mathbf{P}=\begin{pmatrix}2\\-1\\2\end{pmatrix} \tag{1.3.38.4}$$

The following code verifies the intersection:

```
codes/12/q38.py
```

D. Formula-12

1.4.1 Multiplicative Property of Determinants. For any two square matrices $\mathbf{A},\mathbf{B}\in\mathbb{R}^{n\times n}$:

$$\left|\mathbf{AB}\right|=\left|\mathbf{A}\right|\left|\mathbf{B}\right| \tag{1.4.1.1}$$

For an order-2 square matrix $\mathbf{A}=\begin{pmatrix}a&b\\c&d\end{pmatrix}$:

$$\left|\mathbf{A}\right|=ad-bc \tag{1.4.1.2}$$

1.4.2 Derivative of an Exponential Function via Chain Rule:

$$\frac{d}{dx}\left[e^{u(x)}\right]=e^{u(x)}\frac{du}{dx} \tag{1.4.2.1}$$

For $u(x)=kx$:

$$\frac{d}{dx}\left[e^{kx}\right]=ke^{kx} \tag{1.4.2.2}$$

1.4.3 Order and Degree of an Ordinary Differential Equation. For a differential equation relating x,y and derivatives $y^{(k)}=\frac{d^ky}{dx^k}$:

- **Order:** The highest order n of the derivative $\dfrac{d^n y}{dx^n}$ present in the equation.
- **Degree:** The highest power/exponent of the highest-order derivative, defined *only* if the equation can be expressed as a polynomial in all its derivatives. If any derivative occurs within transcendental functions (such as $\log$, $\sin$, e^x), the degree is **not defined**.

1.4.4 Direction Cosines and Direction Ratios of a Line. If a line makes angles α, β, γ with the positive x, y, z axes:

$$l = \cos\alpha, \quad m = \cos\beta, \quad n = \cos\gamma \tag{1.4.4.1}$$

The fundamental identity connecting direction cosines is:

$$l^2 + m^2 + n^2 = 1 \tag{1.4.4.2}$$

If the line makes equal angles with all three coordinate axes ($\alpha = \beta = \gamma$):

$$l = m = n = \pm\frac{1}{\sqrt{3}} \tag{1.4.4.3}$$

1.4.5 Vector and Cartesian Equations of a Straight Line in 3D. A line passing through a point with position vector $\mathbf{a} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$ parallel to direction vector $\mathbf{b} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ has:

$$\text{Vector Form: } \mathbf{r} = \mathbf{a} + \lambda\mathbf{b}, \quad \lambda \in \mathbb{R} \tag{1.4.5.1}$$

$$\text{Cartesian Form: } \frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} = \lambda \tag{1.4.5.2}$$

1.4.6 Power Rule of Indefinite Integration:

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1 \tag{1.4.6.1}$$

1.4.7 Determinant of Powers of a Matrix:

$$\left|\mathbf{A}^k\right| = \left(\left|\mathbf{A}\right|\right)^k, \quad k \in \mathbb{Z}^+ \tag{1.4.7.1}$$

1.4.8 Binary Operations and Associativity. An operation $*$ on a set S is a binary operation if

$$a * b \in S \quad \forall a, b \in S \quad \text{(Closure)} \tag{1.4.8.1}$$

The operation is associative if

$$(a * b) * c = a * (b * c) \quad \forall a, b, c \in S \tag{1.4.8.2}$$

1.4.9 Vector Cross Product and Sine of Angle between Vectors. For non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ with angle $\theta \in [0, \pi]$:

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \, \|\mathbf{b}\| \sin\theta \tag{1.4.9.1}$$

$$\sin\theta = \frac{\|\mathbf{a}\times\mathbf{b}\|}{\|\mathbf{a}\|\,\|\mathbf{b}\|} \tag{1.4.9.2}$$

1.4.10 Scalar Triple Product and Volume of Parallelepiped. The volume of a parallelepiped with coterminous edge vectors $\mathbf{a},\mathbf{b},\mathbf{c}$ is

$$V = |\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})| = \left|\left|\mathbf{a}\quad\mathbf{b}\quad\mathbf{c}\right|\right| \tag{1.4.10.1}$$

1.4.11 Binomial Distribution for Bernoulli Trials. In n independent trials with success probability p and failure probability $q = 1-p$:

$$\Pr(X=k) = \binom{n}{k}p^k q^{n-k}, \quad k\in\{0,1,\ldots,n\} \tag{1.4.11.1}$$

The cumulative distribution satisfies:

$$\Pr(X\le k) = \sum_{r=0}^{k}\binom{n}{r}p^r q^{n-r} = 1-\Pr(X>k) \tag{1.4.11.2}$$

1.4.12 Discrete Probability Distribution Axioms. For a discrete random variable X taking values $\{x_1,x_2,\ldots,x_k\}$:

$$\Pr(X=x_i)\ge 0 \quad \forall i, \qquad \sum_{i=1}^{k}\Pr(X=x_i) = 1 \tag{1.4.12.1}$$

1.4.13 Standard Trigonometric Integrals:

$$\int \sec x\tan x\,dx = \sec x + C \tag{1.4.13.1}$$

$$\int \csc x\cot x\,dx = -\csc x + C \tag{1.4.13.2}$$

1.4.14 Special Integral Form $\int e^x\left[f(x)+f'(x)\right]dx$:

$$\int e^x\left[f(x)+f'(x)\right]dx = e^x f(x) + C \tag{1.4.14.1}$$

Solution: Applying integration by parts to the first term $\int e^x f(x)\,dx$:

$$\int e^x f(x)\,dx = f(x)\int e^x\,dx - \int\left(f'(x)\int e^x\,dx\right)dx \tag{1.4.14.2}$$

$$= f(x)e^x - \int f'(x)e^x\,dx \tag{1.4.14.3}$$

Adding $\int e^x f'(x)\,dx$ to both sides yields (1.4.14.1).

1.4.15 Conditional Probability and Bayes' Inversion:

$$\Pr(A\mid B) = \frac{\Pr(AB)}{\Pr(B)}, \quad \Pr(B\mid A) = \frac{\Pr(AB)}{\Pr(A)} \tag{1.4.15.1}$$

$$\implies \Pr(A \mid B) = \frac{\Pr(B \mid A)\Pr(A)}{\Pr(B)} \tag{1.4.15.2}$$

1.4.16 Separation of Variables for First-Order ODEs:

$$\frac{dy}{dx} = g(x)h(y) \implies \int \frac{1}{h(y)}\,dy = \int g(x)\,dx + C \tag{1.4.16.1}$$

1.4.17 Inverse Trigonometric Identities:

$$\cos^{-1} x = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right), \quad 0 < x \leq 1 \tag{1.4.17.1}$$

$$\tan^{-1} u + \tan^{-1} v = \tan^{-1}\left(\frac{u+v}{1-uv}\right), \quad uv < 1 \tag{1.4.17.2}$$

$$\sin\left(\tan^{-1} w\right) = \frac{w}{\sqrt{1+w^2}} \tag{1.4.17.3}$$

1.4.18 Invariance of Determinants under Elementary Row/Column Operations:

$$\left|\mathbf{A}\right| = \left|\mathbf{EA}\right| \quad \text{for shear operations } R_i \to R_i + kR_j \text{ or } C_i \to C_i + kC_j \tag{1.4.18.1}$$

1.4.19 Equivalence Relation Criteria. A binary relation R on a set A is:

- **Reflexive:** If $(a, a) \in R$ for all $a \in A$.
- **Symmetric:** If $(a, b) \in R \implies (b, a) \in R$.
- **Transitive:** If $(a, b) \in R$ and $(b, c) \in R \implies (a, c) \in R$.

1.4.20 Bijective Maps and Functional Invertibility. A function $f : X \to Y$ is invertible if and only if it is a bijection:

$$\text{Injective: } f(x_1) = f(x_2) \implies x_1 = x_2 \quad \forall x_1, x_2 \in X \tag{1.4.20.1}$$

$$\text{Surjective: } \forall y \in Y,\ \exists x \in X \text{ such that } f(x) = y \tag{1.4.20.2}$$

The inverse function satisfies $f^{-1}(y) = x \iff f(x) = y$.

1.4.21 Logarithmic Differentiation. For variable bases and exponents $u(x)^{v(x)} = w(x)^{z(x)}$:

$$v(x)\ln(u(x)) = z(x)\ln(w(x)) \tag{1.4.21.1}$$

Differentiating implicitly using the product rule and chain rule:

$$\frac{d}{dx}[v \ln u] = \frac{dv}{dx}\ln u + \frac{v}{u}\frac{du}{dx} \tag{1.4.21.2}$$

1.4.22 King's Property of Definite Integrals:

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx \tag{1.4.22.1}$$

Solution: Substitute $t = a - x \implies dx = -dt$. When $x = 0$, $t = a$; when $x = a$, $t = 0$:

$$\int_0^a f(x)\,dx = \int_a^0 f(a-t)\,(-dt) = \int_0^a f(a-t)\,dt = \int_0^a f(a-x)\,dx \tag{1.4.22.2}$$

yielding (1.4.22.1). For symmetric intervals of even functions:

$$\int_{-a}^{a} g(u)\,du = 2\int_{0}^{a} g(u)\,du \quad \text{if } g(-u) = g(u) \tag{1.4.22.3}$$

1.4.23 Partial Fraction Decomposition for Irreducible Quadratic Factors:

$$\frac{P(x)}{(x-\alpha)(x^2+\beta^2)} = \frac{A}{x-\alpha} + \frac{Bx+C}{x^2+\beta^2} \tag{1.4.23.1}$$

Standard primitive integrals:

$$\int \frac{1}{x-\alpha}\,dx = \ln|x-\alpha| + C \tag{1.4.23.2}$$

$$\int \frac{2x}{x^2+\beta^2}\,dx = \ln\left(x^2+\beta^2\right) + C \tag{1.4.23.3}$$

$$\int \frac{1}{x^2+\beta^2}\,dx = \frac{1}{\beta}\tan^{-1}\left(\frac{x}{\beta}\right) + C \tag{1.4.23.4}$$

1.4.24 Homogeneous First-Order Differential Equations:

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right) \tag{1.4.24.1}$$

Solution: Substituting $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$:

$$v + x\frac{dv}{dx} = F(v) \implies \frac{dv}{F(v)-v} = \frac{dx}{x} \tag{1.4.24.2}$$

reducing the equation to separable variables.

1.4.25 First-Order Linear Differential Equations:

$$\frac{dy}{dx} + P(x)y = Q(x) \tag{1.4.25.1}$$

Solution: Multiplying by the integrating factor $\mu(x) = e^{\int P(x)\,dx}$:

$$\frac{d}{dx}\left[y\cdot\mu(x)\right] = Q(x)\mu(x) \tag{1.4.25.2}$$

$$\implies y\cdot e^{\int P(x)\,dx} = \int Q(x)e^{\int P(x)\,dx}\,dx + C \tag{1.4.25.3}$$

1.4.26 Unit Vector and Orthogonal Vector Projection:

$$\hat{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|}, \quad \|\hat{u}\| = 1 \tag{1.4.26.1}$$

For vectors $\mathbf{a}$ and $\mathbf{b}$:

$$\mathbf{a}\cdot\mathbf{b} = a_x b_x + a_y b_y + a_z b_z \tag{1.4.26.2}$$

1.4.27 Related Rates of Change via Derivative Chains. If $V(t) = f(x(t))$ and $S(t) =$

$g(x(t))$:

$$\frac{dV}{dt} = f'(x)\frac{dx}{dt}, \quad \frac{dS}{dt} = g'(x)\frac{dx}{dt} \implies \frac{dS}{dt} = \frac{g'(x)}{f'(x)}\frac{dV}{dt} \tag{1.4.27.1}$$

For a cube with side length x:

$$V = x^3 \implies \frac{dV}{dt} = 3x^2\frac{dx}{dt}, \quad S = 6x^2 \implies \frac{dS}{dt} = 12x\frac{dx}{dt} \tag{1.4.27.2}$$

1.4.28 Plane Passing Through Three Non-Collinear Points. The Cartesian equation of a plane through $\mathbf{a} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} x_3 \\ y_3 \\ z_3 \end{pmatrix}$ is

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0 \tag{1.4.28.1}$$

In vector form with normal $\mathbf{n} = (\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a})$:

$$\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n} \tag{1.4.28.2}$$

1.4.29 Calculus-Based Extremum and Optimization. For a twice-differentiable functional $Z(y)$:

$$\text{Stationary Condition: } \frac{dZ}{dy} = 0 \tag{1.4.29.1}$$

$$\text{Local Minimum: } \left.\frac{d^2Z}{dy^2}\right|_{y=y^*} > 0 \tag{1.4.29.2}$$

1.4.30 Definite Integral as the Limit of Riemann Sums:

$$\int_a^b f(x)\,dx = \lim_{h\to 0} h\sum_{r=1}^{n} f(a + rh), \quad \text{where } nh = b - a \tag{1.4.30.1}$$

Fundamental summation formulas:

$$\sum_{r=1}^{n} r = \frac{n(n+1)}{2} \tag{1.4.30.2}$$

$$\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6} \tag{1.4.30.3}$$

$$\sum_{r=1}^{n} q^r = q\frac{q^n - 1}{q - 1}, \quad q \neq 1 \tag{1.4.30.4}$$

Asymptotic exponential limit:

$$\lim_{h\to 0} \frac{e^{kh} - 1}{h} = k \tag{1.4.30.5}$$

1.4.31 Area of Plane Region Bounded by Curves:

$$A = \int_a^b \left(y_{\text{upper}}(x) - y_{\text{lower}}(x)\right)\, dx \tag{1.4.31.1}$$

1.4.32 Matrix Inverse via Adjugate Matrix:

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|}\operatorname{adj}(\mathbf{A}) = \frac{1}{|\mathbf{A}|}\mathbf{C}^{\top} \tag{1.4.32.1}$$

where $C_{ij} = (-1)^{i+j} M_{ij}$ is the cofactor of entry a_{ij}. The unique solution to linear system $\mathbf{Ax} = \mathbf{b}$ is:

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} \tag{1.4.32.2}$$

1.4.33 Matrix Inversion by Gauss-Jordan Row Elimination:

$$[\mathbf{A} \mid \mathbf{I}] \xrightarrow{R_i \to cR_i,\, R_i \leftrightarrow R_j,\, R_i \to R_i + kR_j} \left[\mathbf{I} \mid \mathbf{A}^{-1}\right] \tag{1.4.33.1}$$

1.4.34 Linear Programming Problem (LPP) and Fundamental Corner-Point Theorem. For the canonical problem:

$$\text{Maximize/Minimize: } Z = \mathbf{c}^{\top}\mathbf{x} \tag{1.4.34.1}$$

$$\text{Subject to: } \mathbf{Ax} \le \mathbf{b}, \quad \mathbf{x} \ge \mathbf{0} \tag{1.4.34.2}$$

The optimal value of the objective function Z over a bounded convex polygonal feasible region occurs at one of the extreme (corner) vertices of the region.

1.4.35 Law of Total Probability and Bayes' Theorem. If $\{E_1, E_2, \ldots, E_n\}$ forms a partition of sample space S:

$$\Pr(A) = \sum_{j=1}^{n} \Pr(E_j)\Pr(A \mid E_j) \tag{1.4.35.1}$$

$$\Pr(E_k \mid A) = \frac{\Pr(E_k)\Pr(A \mid E_k)}{\sum_{j=1}^{n} \Pr(E_j)\Pr(A \mid E_j)} \tag{1.4.35.2}$$

1.4.36 Distance Between Parallel Planes. For two parallel planes $\Pi_1 : Ax+By+Cz+D_1 = 0$ and $\Pi_2 : Ax + By + Cz + D_2 = 0$:

$$D = \frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}} = \frac{|d_1 - d_2|}{\|\mathbf{n}\|} \tag{1.4.36.1}$$

1.4.37 Intersection of a Line and a Plane in 3D. For a line $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$ and plane $\mathbf{r} \cdot \mathbf{n} = d$:

$$(\mathbf{a} + \lambda\mathbf{b}) \cdot \mathbf{n} = d \implies \lambda = \frac{d - \mathbf{a}\cdot\mathbf{n}}{\mathbf{b}\cdot\mathbf{n}}, \quad \mathbf{b}\cdot\mathbf{n} \neq 0 \tag{1.4.37.1}$$

Substituting λ back into the line equation yields the coordinates of the unique point of intersection.